Conditioning results for the 2024 Hake Reference Set

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Summary

Results are presented for the conditioning of the 2024 hake Reference Set (RS) of nine Operating Models (OMs). On the whole, the updated RS is reasonably consistent with the last RS from 2022, barring two runs where a more depleted *M. capensis* stock is estimated – the remaining OMs estimate both stocks to be above B_{MSY}. There seems to be some multi-modality in the negative log-likelihood surface, which is not uncommon for the hake models, indicating that the data are not able to distinguish between two or more possible outcomes. In particular, the *M. capensis* depletion levels seem not to be very well determined. An exercise will be undertaken whereby runs for which the *M. capensis* is estimated to be more depleted are projected forward under the rules of OMP-2022 to ascertain whether the OMP secures *M. capensis* recovery for such cases.

Keywords: OMP-2022, Reference Set conditioning, South African hake

Note that this document is a revision of the one presented to the DWG on 8th October 2024, with the addition of Appendix B as well as the correction of glitches in some tables.

Introduction

The Reference Set (RS) of Operating Models (OMs) is updated with the latest survey and commercial data. The last update of the entire RS was in 2022 (Ross-Gillespie, 2022), and the last update of the central Reference Case (RC) OM was in 2023 (Ross-Gillespie and Butterworth, 2023).

The RS consists of nine models. There are three options for the central year in the shift from predominantly *M. capensis* in the historical catch (for which no species information is available) to predominantly *M. paradoxus.*

1. Centre of the shift occurred in 1952.
2. Centre of the shift occurred in 1958.
3. Centre of the shift occurred in 1963.

Additionally, three options for the form of the stock-recruitment function have been considered.

1. Modified Ricker[2]
2. Beverton-Holt (B-H) with *h* fixed at 0.90
3. Beverton-Holt (B-H) with *h* fixed at 0.70

Each of the runs for which results have been reported in this document have undergone "jittering" whereby the starting parameters are randomly jittered by a percentage and the minimisation is restarted to try to ensure as best as possible that a global minimum has been found. This jittering process was repeated at least 50 times for each RS OM.

Data updates

The commercial catch data have been updated to include 2023 catches, as well as come correction to catches for 2010-2022, and are listed in Table 1. The GLM CPUE series have also been updated to 2023 and are listed in Table 2. As can be seen in Figure 1, the 2024 GLM CPUE series revision is generally consistent with the 2023 series. Table 3a and b list the survey abundance estimates including the new 2024 estimates.

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[2] The modified Ricker equation is given by $R_y^g = \alpha B_y^{♀,sp} \exp\left(-\beta\left(B_y^{♀,sp}\right)^\gamma\right) e^{(\varsigma_y - \sigma_R^2/2)}$ with $\alpha = R_0 \exp\left(\beta\left(K^{♀,sp}\right)^\gamma\right) / K^{♀,sp}$ and $\beta = \dfrac{\ln(5h)}{\left(K^{♀,sp}\right)^\gamma\left(1-5^{-\gamma}\right)}$.

Results

Table 4 lists key parameter estimates for the updated RS, while Table 5 lists the negative log-likelihood components.

Figure 2a-c plot the spawning biomass trajectories of the 2024 RS alongside those from the 2022 RS (which was the last time the whole RS was updated). Figure 3 plots the spawning biomass for the 2022, 2023 and 2024 RC, while Figure 4 plots recruitment related output and Figure 5 the CPUE fits for these three RC OMs.

More detailed results for the RS OMs can be found in Appendix B.

Discussion

Some discussion points are listed below.

Reference set

- Barring two cases, Figure 2a-c indicate that the updated RS is fairly consistent with the 2022 RS. The two exceptions are the following.
    - An alternative maximum likelihood estimate (MLE) was found for RS01 during the jittering process, which has a similar negative log-likelihood but estimates a much more depleted *M. capensis* stock. More details for this run are provided in Appendix A. This alternative run has been excluded from the results shown in the main text because its estimate for the *M. paradoxus* B_{MSY}/K^{sp} is low at 0.14. This value, however, is not much lower than for the RS01 run reported on in the main text (0.16) and there seem little grounds to choose between the two as it seems that the data are not able to distinguish between these two possible realities.
    - The best estimate for RS07 also indicates a much more depleted *M. capensis* stock than the other OMs. It is not impossible that another mode where the *M. capensis* stock is well recovered exists but has not yet been found through the jittering process.
    - Note that multi-modality in the negative log-likelihood surface is not an uncommon occurrence for these hake models. Given the apparent plausibility of a more depleted *M. capensis* stock for at least RS01 and RS07, an exercise will be undertaken whereby these OMs are projected forward under the rules of OMP-2022 to assess whether the OMP is able secure *M capensis* recovery in such a situation should it in fact be the reality.
    - The B_{MSY}/K^{sp} estimates for the Beverton-Holt OMs with $h = 0.90$ are low (see Table 4). These OMs could be re-run with a penalty to try force $B_{MSY}/K^{sp} > 0.15$.
- The B_{MSY}/K^{sp}estimates for the Beverton-Holt OMs with $h = 0.90$ (RS04-RS06) are low for *M. paradoxus* (0.13-0.14). These could be re-run with a penalty to restrict B_{MSY}/K^{sp} >0.15.

Reference case

- Overall, however, the RC results are reasonably consistent with what they were in 2023, with both stocks only slightly more depleted in the update than for the 2023 RC (see Table 4).

Acknowledgements

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Table 1: Species-disaggregated catches (in thousand tons) by fleet of South African hake from the south and west coasts for the period 1978-present, as provided by D. Durholtz (*pers. comm.*). Grey highlighted cells indicate updated catch values.

	M. paradoxus				*M. capensis*					
	Offshore		Longline		Offshore		Inshore	Longline		Handline
	WC	SC	WC	SC	WC	SC	SC	WC	SC	SC
1978	101.042	3.220	-	-	26.470	4.365	4.931	-	-	-
1979	94.331	1.924	-	-	39.192	4.995	6.093	-	-	-
1980	99.654	2.206	-	-	33.873	4.254	9.121	-	-	-
1981	88.883	0.910	-	-	32.048	4.575	9.400	-	-	-
1982	83.618	3.353	-	-	29.732	8.005	8.089	-	-	-
1983	71.238	4.723	0.126	-	23.195	7.792	7.672	0.104	-	-
1984	82.358	3.796	0.200	0.005	28.897	7.139	9.035	0.166	0.011	-
1985	94.428	8.059	0.638	0.091	30.642	11.957	9.203	0.529	0.201	0.065
1986	103.756	8.580	0.753	0.094	30.049	7.385	8.724	0.625	0.208	0.084
1987	93.517	7.460	1.952	0.110	24.008	8.225	8.607	1.619	0.243	0.096
1988	79.913	5.876	2.833	0.103	26.669	8.640	8.417	2.350	0.228	0.071
1989	82.230	6.182	0.158	0.010	25.029	12.730	10.038	0.132	0.022	0.137
1990	81.996	9.341	0.211	-	21.640	13.451	10.012	0.175	-	0.348
1991	87.093	12.448	-	0.932	19.357	9.626	8.206	-	2.068	1.270
1992	84.768	17.297	-	0.466	18.519	9.165	9.252	-	1.034	1.099
1993	102.125	9.880	-	-	15.940	4.380	8.870	-	-	0.278
1994	103.541	6.726	0.882	0.194	20.327	4.326	9.569	0.732	0.432	0.449
1995	100.268	4.004	0.523	0.202	20.629	3.146	10.630	0.434	0.448	0.756
1996	107.381	8.966	1.308	0.568	21.794	4.323	11.062	1.086	1.260	1.515
1997	100.654	10.509	1.410	0.582	16.500	5.327	8.834	1.170	1.290	1.404
1998	111.154	9.742	0.505	0.457	16.499	4.411	8.283	0.419	1.014	1.738
1999	88.581	11.420	1.532	1.288	15.179	3.926	8.595	1.272	2.856	2.749
2000	96.587	7.700	2.706	3.105	21.114	5.830	10.906	2.000	1.977	5.500
2001	101.247	7.850	1.417	0.084	16.349	8.306	11.836	2.394	1.527	7.300
2002	91.207	12.443	4.469	1.585	13.724	6.141	9.581	2.391	2.546	3.500
2003	93.711	17.397	3.305	1.252	11.665	7.636	9.883	2.526	3.078	3.000
2004	85.722	26.065	2.855	1.196	12.510	8.704	10.004	2.297	2.731	1.600
2005	85.869	21.778	3.091	0.472	9.398	7.468	7.881	2.773	3.270	0.700
2006	81.513	18.050	3.241	0.485	11.984	6.578	5.524	2.520	3.227	0.400
2007	92.724	13.488	2.512	3.021	16.145	3.757	6.350	2.522	2.522	0.400
2008	85.538	13.191	2.255	0.809	13.838	4.316	5.496	1.937	1.893	0.231
2009	68.202	10.895	2.410	1.069	12.296	4.806	5.639	2.828	2.520	0.265
2010	69.459	15.695	2.394	1.527	10.193	4.124	5.965	3.086	3.024	0.275
2011	75.697	18.580	2.522	0.140	15.639	4.240	6.437	3.521	3.047	0.186
2012	80.978	16.687	4.358	0.306	12.986	4.614	3.423	2.570	1.737	0.008
2013	75.005	29.155	6.056	0.060	8.965	4.503	2.920	2.606	1.308	0.000
2014	74.619	40.308	6.879	0.008	9.970	6.159	2.965	2.123	0.315	0.001
2015	79.639	30.858	5.205	0.023	13.431	3.924	3.082	3.025	0.069	0.001
2016	93.408	19.799	3.697	0.001	15.091	2.922	4.182	5.745	0.003	0.001
2017	72.406	30.878	5.300	0.025	15.646	4.468	2.813	2.813	0.126	0.004
2018	65.657	29.022	5.217	0.090	12.537	11.811	3.985	2.646	0.487	0.024
2019	75.446	21.979	5.328	0.034	14.127	8.840	4.743	3.623	0.299	0.009
2020	101.030	10.809	5.847	0.047	15.916	2.990	4.576	2.348	0.321	0.004
2021	95.635	12.317	5.892	0.018	22.276	8.098	5.439	2.932	0.194	0.010
2022	75.277	13.691	4.970	0.019	13.734	3.736	5.402	2.933	0.080	0.001
2023	71.756	14.747	5.049	0.007	12.774	3.610	3.700	2.366	0.033	0.007

Table 2: GLM standardized CPUE data for *M. paradoxus* and *M. capensis*, corresponding to the Model A6b species splitting algorithm provided by D. Durholtz (*pers. comm.*). These series have been plotted in Figure 1.

	GLM CPUE (kg min^{-1})			
	M. paradoxus		*M. capensis*	
Year	West Coast	South Coast	West Coast	South Coast
1978	1.177	0.554	0.862	0.743
1979	1.184	0.475	1.183	0.832
1980	1.112	0.777	1.065	1.004
1981	1.090	0.535	1.108	0.873
1982	1.067	0.746	0.965	0.919
1983	1.176	0.750	1.193	1.078
1984	1.202	0.819	1.214	1.250
1985	1.403	1.062	1.358	1.571
1986	1.221	1.074	1.130	1.297
1987	1.028	0.994	0.956	1.190
1988	1.000	0.824	0.876	1.216
1989	1.079	0.854	0.934	1.351
1990	1.139	1.086	0.836	1.620
1991	1.269	1.189	1.017	1.451
1992	1.135	1.315	1.154	1.363
1993	1.147	1.115	1.117	1.009
1994	1.222	1.004	1.308	1.210
1995	1.091	0.780	1.256	1.111
1996	1.202	1.102	1.442	1.112
1997	1.069	1.182	1.225	0.940
1998	1.208	1.120	1.390	0.926
1999	1.003	1.302	1.244	1.003
2000	0.913	1.020	1.172	1.035
2001	0.738	1.009	0.838	0.838
2002	0.721	0.919	0.800	0.899
2003	0.826	1.051	0.721	0.955
2004	0.689	0.904	0.675	0.803
2005	0.671	0.790	0.517	0.692
2006	0.706	0.817	0.587	0.550
2007	0.848	0.846	0.613	0.539
2008	0.919	0.869	0.748	0.699
2009	0.947	1.009	1.080	1.196
2010	1.054	1.108	0.969	0.977
2011	0.831	1.253	0.582	1.120
2012	0.909	1.118	0.975	0.764
2013	0.792	1.240	0.779	0.757
2014	0.914	1.173	0.825	0.548
2015	0.980	1.109	0.794	0.590
2016	1.005	1.149	0.815	0.757
2017	1.070	1.271	1.032	0.982
2018	0.946	1.408	1.063	1.724
2019	0.863	1.102	1.036	1.287
2020	0.965	1.098	1.176	0.915
2021	0.979	1.163	1.380	0.848
2022	0.777	1.084	1.100	0.710
2023	0.715	0.830	0.890	0.747

Table 3a: Survey abundance estimates and associated standard errors in thousand tons for ***M. paradoxus*** for the depth range 0-500m for the South Coast and for the West Coast (Fairweather, 2024). Note that the 2022 and 2023 surveys were conducted using the FV Compass Challenger and the values listed in this table (in bold italics) were scaled by the calibration factor of 1.25 in the OMs to account for different catchability coefficients.

	West coast				South coast			
Year	Summer		Winter		Spring (Sept)		Autumn (Apr/May)	
	Biomass	(s.e.)	Biomass	(s.e.)	Biomass	(s.e.)	Biomass	(s.e.)
1985	168.989	(37.765)	290.281	(63.295)	-	-	-	-
1986	202.334	(37.745)	147.378	(21.667)	11.280	(3.111)	-	-
1987	284.434	(54.165)	180.158	(39.047)	16.381	(3.033)	-	-
1988	138.534	(20.303)	252.121	(71.246)	-	-	28.293	(8.673)
1989	-	-	434.092	(142.716)	-	-	-	-
1990	307.615	(87.841)	205.704	(43.607)	-	-	-	-
1991	331.177	(81.633)	-	-	-	-	27.570	(8.153)
1992	225.755	(33.711)	-	-	-	-	25.036	(6.650)
1993	340.079	(51.427)	-	-	-	-	162.375	(81.691)
1994	333.499	(56.259)	-	-	-	-	108.179	(38.369)
1995	317.104	(76.709)	-	-	-	-	70.890	(39.330)
1996	474.270	(92.744)	-	-	-	-	68.859	(19.929)
1997	543.615	(96.043)	-	-	-	-	121.707	(51.507)
1998	-	-	-	-	-	-	-	-
1999	542.830	(110.541)	-	-	-	-	263.256	(59.439)
2000	-	-	-	-	-	-	-	-
2001	-	-	-	-	16.668	(7.159)	-	-
2002	251.820	(32.690)	-	-	-	-	-	-
2003	386.321	(63.565)	-	-	98.434	(42.249)	185.345	(82.188)
2004	271.540	(55.710)	-	-	70.001	(22.156)	39.822	(22.153)
2005	296.065	(42.409)	-	-	-	-	26.691	(6.017)
2006	316.247	(57.332)	-	-	68.507	(18.283)	34.868	(5.843)
2007	407.377	(77.222)	-	-	66.267	(21.966)	102.195	(53.688)
2008	238.143	(37.018)	-	-	25.661	(8.324)	33.034	(9.340)
2009	310.760	(27.768)	-	-	-	-	45.030	(15.551)
2010	576.848	(88.202)	-	-	-	-	46.938	(12.160)
2011	380.185	(128.013)	-	-	-	-	21.054	(6.531)
2012	405.865	(59.099)	-	-	-	-	-	-
2013	136.260	(25.116)	-	-	-	-	-	-
2014	269.482	(37.492)	-	-	-	-	62.925	(24.802)
2015	207.583	(24.057)	-	-	-	-	111.411	(51.852)
2016	312.876	(33.250)	-	-	-	-	94.177	(51.731)
2017	319.024	(58.766)	-	-	-	-	-	-
2018	-	-	-	-	-	-	-	-
2019	243.560	(51.558)	-	-	-	-	33.176	(15.444)
2020	243.090	(43.989)	-	-	-	-	-	-
2021	-	-	-	-	-	-	50.060	(20.430)
2022	***254.715***	***(68.289)***	-	-	-	-	-	-
2023	***102.060***	***(13.877)***					***16.327***	***(3.945)***
2024	240.531	(50.557)					72.453	(25.329)

Table 3b: Survey abundance estimates and associated standard errors in thousand tons for ***M. capensis*** for the depth range 0-500m for the South Coast and for the West Coast (Fairweather, 2024). The 2022 and 2023 surveys were conducted using the FV Compass Challenger and the values listed in this table (in bold italics) were scaled by the calibration factor of 1.25 in the OMs to account for different catchability coefficients.

	West coast				South coast			
Year	Summer		Winter		Spring (Sept)		Autumn (Apr/May)	
	Biomass	(s.e.)	Biomass	(s.e.)	Biomass	(s.e.)	Biomass	(s.e.)
1985	102.929	(18.888)	159.198	(18.982)	-	-	-	-
1986	113.154	(23.474)	115.218	(19.733)	96.768	(10.737)	-	-
1987	75.438	(9.709)	83.050	(10.306)	137.008	(13.057)	-	-
1988	66.365	(9.930)	48.046	(9.574)	-	-	154.548	(23.984)
1989	-	-	294.740	(67.495)	-	-	-	-
1990	400.142	(97.102)	156.337	(22.507)	-	-	-	-
1991	67.565	(9.656)	-	-	-	-	276.607	(25.274)
1992	95.401	(11.892)	-	-	-	-	124.495	(13.600)
1993	93.613	(14.390)	-	-	-	-	144.551	(12.379)
1994	124.497	(37.845)	-	-	-	-	153.790	(20.310)
1995	193.292	(24.270)	-	-	-	-	222.464	(31.245)
1996	87.969	(9.866)	-	-	-	-	222.176	(23.144)
1997	252.606	(42.721)	-	-	-	-	163.163	(17.274)
1998	-	-	-	-	-	-	-	-
1999	188.624	(31.362)	-	-	-	-	171.946	(13.330)
2000	-	-	-	-	-	-	-	-
2001	-	-	-	-	117.590	(20.093)	-	-
2002	105.093	(16.130)	-	-	-	-	-	-
2003	73.020	(12.518)	-	-	73.604	(9.142)	117.538	(17.192)
2004	194.294	(30.714)	-	-	96.933	(13.936)	92.796	(11.318)
2005	63.363	(11.498)	-	-	-	-	68.672	(5.302)
2006	73.655	(17.255)	-	-	92.831	(8.998)	116.298	(11.931)
2007	73.230	(9.306)	-	-	67.937	(6.553)	65.935	(5.303)
2008	52.577	(7.069)	-	-	87.836	(9.723)	102.169	(9.681)
2009	140.437	(26.486)	-	-	-	-	111.191	(10.832)
2010	162.402	(34.891)	-	-	-	-	170.261	(33.235)
2011	89.095	(23.574)	-	-	-	-	105.424	(10.688)
2012	84.746	(8.331)	-	-	-	-	-	-
2013	30.383	(4.575)	-	-	-	-	-	-
2014	219.756	(60.342)	-	-	-	-	63.389	(6.415)
2015	65.086	(9.178)	-	-	-	-	76.059	(6.873)
2016	115.058	(30.400)	-	-	-	-	83.197	(6.600)
2017	69.289	(14.486)	-	-	-	-	-	-
2018	-	-	-	-	-	-	-	-
2019	62.560	(7.697)	-	-	-	-	132.099	(14.486)
2020	109.983	(11.836)	-	-	-	-	-	-
2021	-	-	-	-	-	-	89.119	(10.798)
2022	***140.573***	***(29.659)***	**-**	**-**	**-**	**-**	**-**	**-**
2023	***184.518***	***(55.745)***					***87.933***	***(12.970)***
2024	295.141	(66.385)					114.403	(15.951)

Table 4: Key parameter estimates for the updated RS models (biomass units are thousand tons).

(a) *M. paradoxus*

Model name	Central Year	Stock Recruit	K^{sp}	B^{sp}_{MSY}	B^{sp}_{2024}	B^{tot}_{2024}	B^{sp}_{2023}/K^{sp}	B^{sp}_{2024}/K^{sp}	$B^{sp}_{2023}/B^{sp}_{MSY}$	$B^{sp}_{2024}/B^{sp}_{MSY}$	B^{sp}_{MSY}/K^{sp}	MSY
(0) 2023 RC	1958	Ricker	470	72	-	-	0.24	-	1.59	-	0.15	135
(1) RS01	1952		510	80	101	343	0.21	0.20	1.36	1.25	0.16	130
(2) RS02 RC	1958	Ricker	497	79	101	342	0.22	0.20	1.39	1.29	0.16	130
(3) RS03	1963		418	87	118	376	0.30	0.28	1.46	1.36	0.21	131
(4) RS04	1952		981	140	177	441	0.20	0.18	1.38	1.26	0.14	117
(5) RS05	1958	Beverton-Holt (h=0.9)	978	136	171	447	0.19	0.18	1.36	1.26	0.14	118
(6) RS06	1963		959	129	201	529	0.22	0.21	1.65	1.56	0.13	121
(7) RS07	1952		1404	340	420	927	0.31	0.30	1.29	1.24	0.24	114
(8) RS08	1958	Beverton-Holt (h=0.7)	1382	335	401	889	0.30	0.29	1.25	1.20	0.24	113
(9) RS09	1963		1482	357	545	1181	0.38	0.37	1.58	1.53	0.24	120

(b) *M. capensis*

Model name	Central Year	Stock Recruit	K^{sp}	B^{sp}_{MSY}	B^{sp}_{2024}	B^{tot}_{2024}	B^{sp}_{2023}/K^{sp}	B^{sp}_{2024}/K^{sp}	$B^{sp}_{2023}/B^{sp}_{MSY}$	$B^{sp}_{2024}/B^{sp}_{MSY}$	B^{sp}_{MSY}/K^{sp}	MSY
(0) 2023 RC	1958	Ricker	237	70	NA	NA	0.83	NA	2.82	NA	0.30	83
(1) RS01	1952		342	76	279	799	0.79	0.82	3.58	3.69	0.22	109
(2) RS02 RC	1958	Ricker	227	59	175	530	0.74	0.77	2.84	2.95	0.26	84
(3) RS03	1963		386	117	327	921	0.82	0.85	2.71	2.80	0.30	103
(4) RS04	1952		1090	168	926	2491	0.82	0.85	5.35	5.52	0.15	162
(5) RS05	1958	Beverton-Holt (h=0.9)	1062	164	897	2419	0.82	0.85	5.31	5.48	0.15	157
(6) RS06	1963		1185	182	1021	2738	0.84	0.86	5.43	5.60	0.15	176
(7) RS07	1952		411	110	66	267	0.15	0.16	0.55	0.60	0.27	49
(8) RS08	1958	Beverton-Holt (h=0.7)	1363	352	1166	3110	0.83	0.86	3.22	3.31	0.26	147
(9) RS09	1963		1685	435	1486	3935	0.86	0.88	3.32	3.42	0.26	181

Table 5a: Negative log-likelihood components are shown for the updated RS models. For the updated RS OMs (which use the Baranov formulation for the catch equation), the penalty column corresponds to the catch penalty for when the model-predicted catches do not match the observed catches perfectly. The total in the last column excludes this penalty.

Run	GLM CPUE	ICSEAF CPUE	Survey abun.	Comm. CAL	Survey CAL	Recruit. resid.	ALKs	Penalties	Total (w/o pen.)
RS01 (Ricker, CY=1952)	−252.14	−38.07	−40.19	−1615.60	−1806.82	11.39	129.00	0.01	−3612.43
RS02 (Ricker, CY=1958)	−252.79	−39.59	−40.49	−1616.51	−1805.61	12.28	128.78	0.02	−3613.93
RS03 (Ricker, CY=1963)	−253.14	−37.96	−39.97	−1614.36	−1807.71	11.00	129.21	0.14	−3612.94
RS04 (B–H, h=0.90, 1952)	−238.81	−37.12	−40.37	−1608.49	−1810.86	20.22	130.75	0.00	−3584.67
RS05 (B–H, h=0.90, 1958)	−222.46	−33.49	−39.30	−1618.96	−1806.76	15.31	129.12	0.08	−3576.53
RS06 (B–H, h=0.90, 1963)	−215.87	−34.74	−37.46	−1620.82	−1807.74	13.04	129.69	0.05	−3573.89
RS07 (B–H, h=0.70, 1952)	−223.76	−29.59	−39.63	−1622.99	−1804.29	11.55	129.26	0.04	−3579.45
RS08 (B–H, h=0.70, 1958)	−216.86	−24.43	−37.59	−1622.65	−1810.53	13.34	130.34	0.05	−3568.38
RS09 (B–H, h=0.70, 1963)	−216.40	−28.74	−37.08	−1622.90	−1810.43	12.45	130.58	0.04	−3572.53

Table 5b: Negative log-likelihoods expressed relative to the updated Reference Case OM (RS02). Cells highlighted in grey indicate a worse negative log-likelihood.

Run	GLM CPUE	ICSEAF CPUE	Survey abun.	Comm. CAL	Survey CAL	Recruit. resid.	ALKs	Penalties	Total (w/o pen.)
RS01 (Ricker, CY=1952)	0.66	1.52	0.30	0.92	−1.21	−0.89	0.21	−0.01	1.50
RS02 (Ricker, CY=1958)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
RS03 (Ricker, CY=1963)	−0.35	1.63	0.52	2.15	−2.10	−1.28	0.42	0.13	0.98
RS04 (B–H, h=0.90, 1952)	13.98	2.47	0.12	8.02	−5.25	7.94	1.97	−0.01	29.25
RS05 (B–H, h=0.90, 1958)	30.33	6.11	1.19	−2.45	−1.15	3.03	0.34	0.07	37.39
RS06 (B–H, h=0.90, 1963)	36.92	4.85	3.03	−4.31	−2.13	0.76	0.91	0.03	40.04
RS07 (B–H, h=0.70, 1952)	29.03	10.01	0.86	−6.48	1.32	−0.73	0.48	0.02	34.48
RS08 (B–H, h=0.70, 1958)	35.93	15.17	2.90	−6.14	−4.93	1.06	1.56	0.04	45.54
RS09 (B–H, h=0.70, 1963)	36.39	10.85	3.40	−6.39	−4.82	0.17	1.80	0.02	41.40

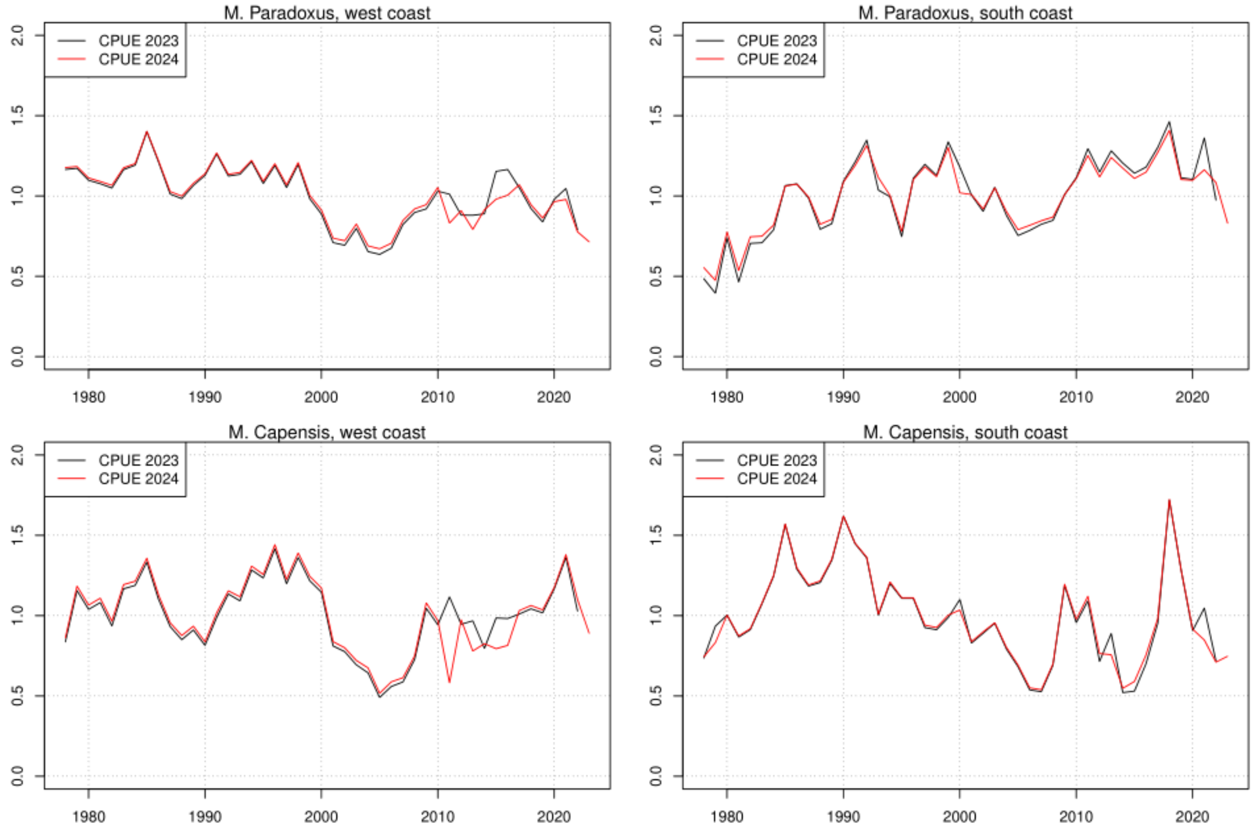


Figure 1: Plot of the 2024 commercial CPUE series (red lines), superimposed onto the 2023 CPUE series (black lines), which was normalised so that it has the same mean as the 2024 series.

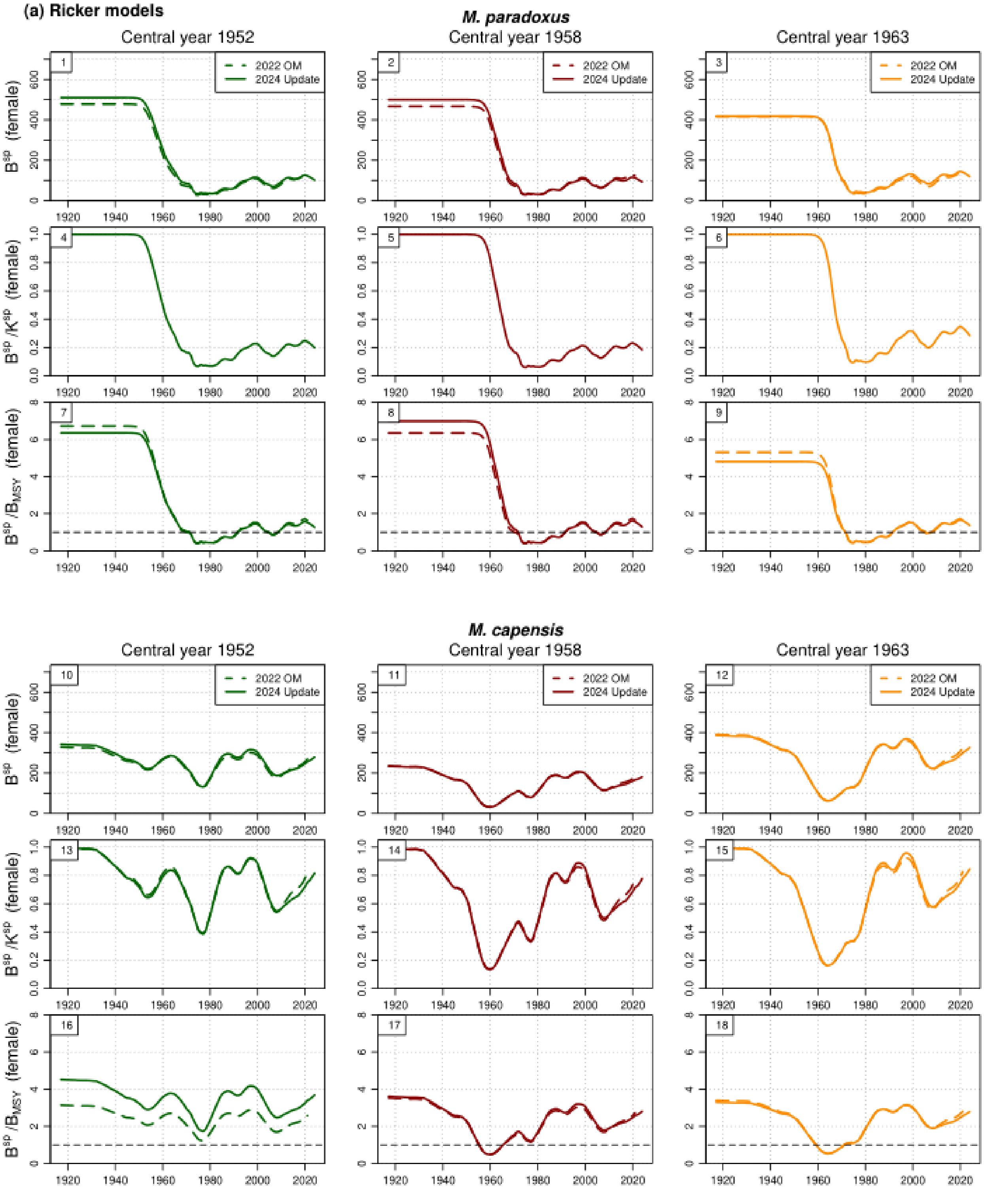


Figure 2a: Spawning biomass for the three **Ricker models**, shown for the 2022 RS OMs and the 2024 update. For each species, the top row shows spawning biomass in absolute terms, the second row relative to pristine spawning biomass and the third row relative to B_{MSY}.

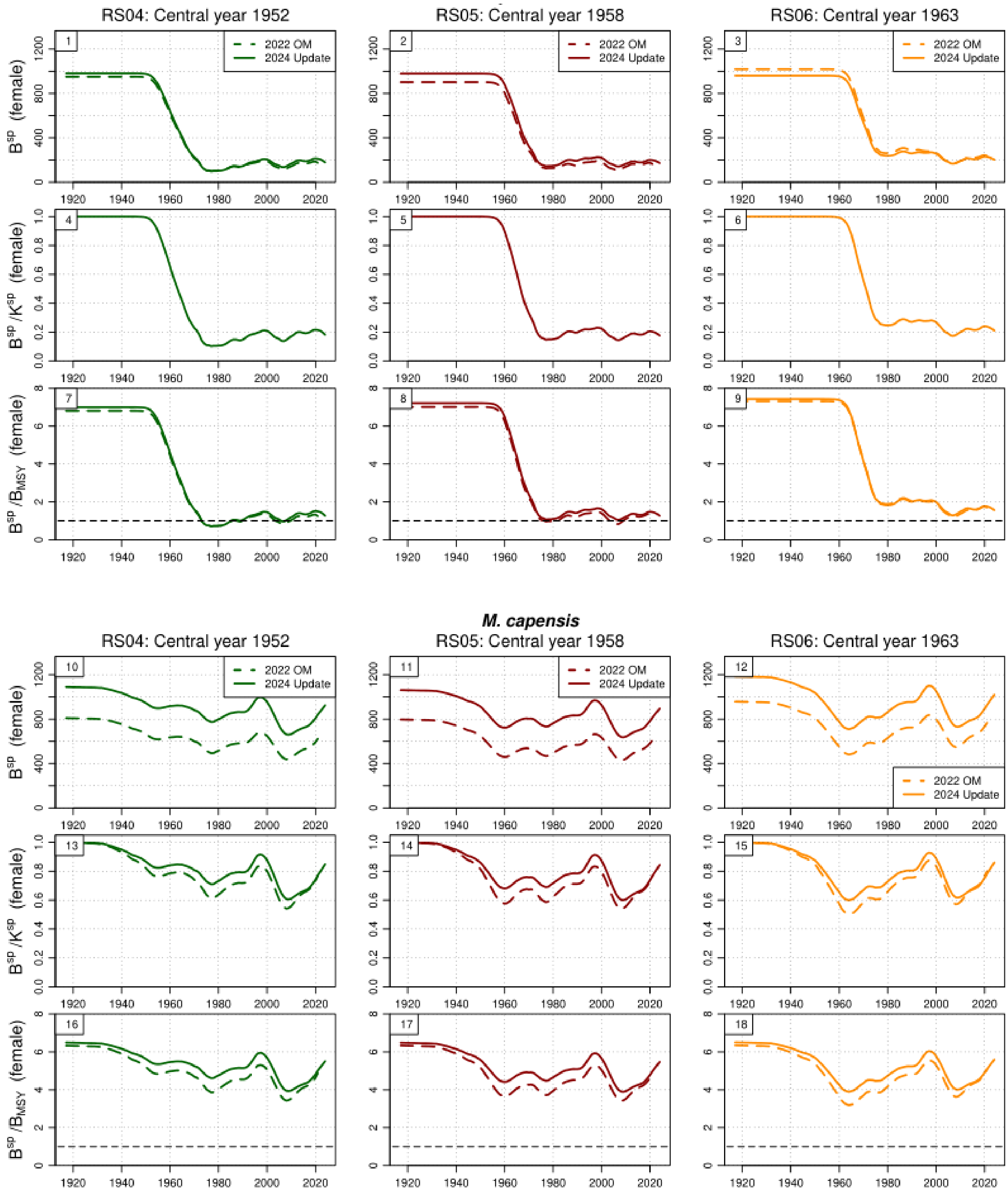


Figure 2b: Repeat of Figure 2a, but for the **Beverton-Holt models** with **h=0.90**.

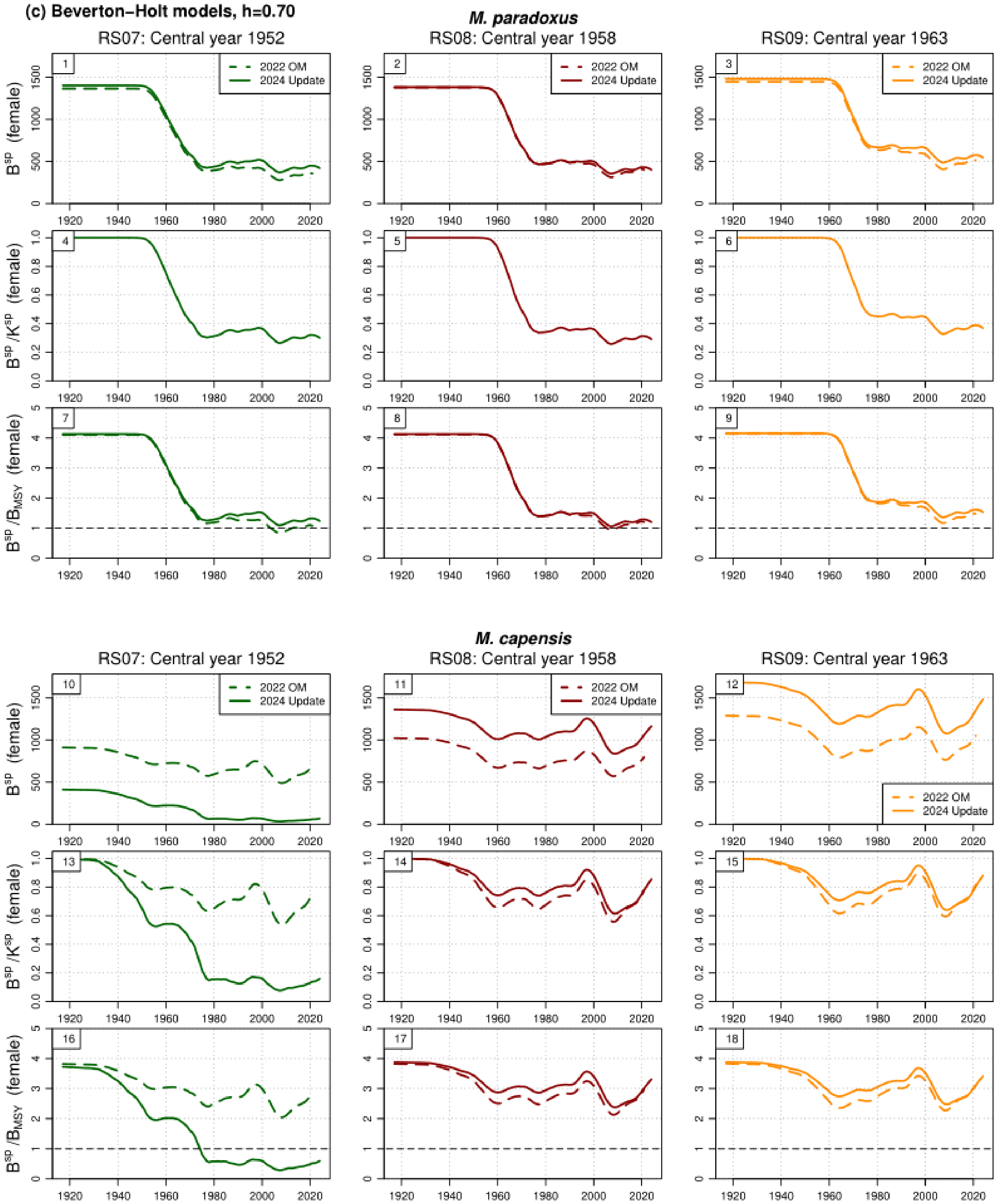


Figure 2c: Repeat of Figure 2b, but for the **Beverton-Holt models** with **h=0.70**.

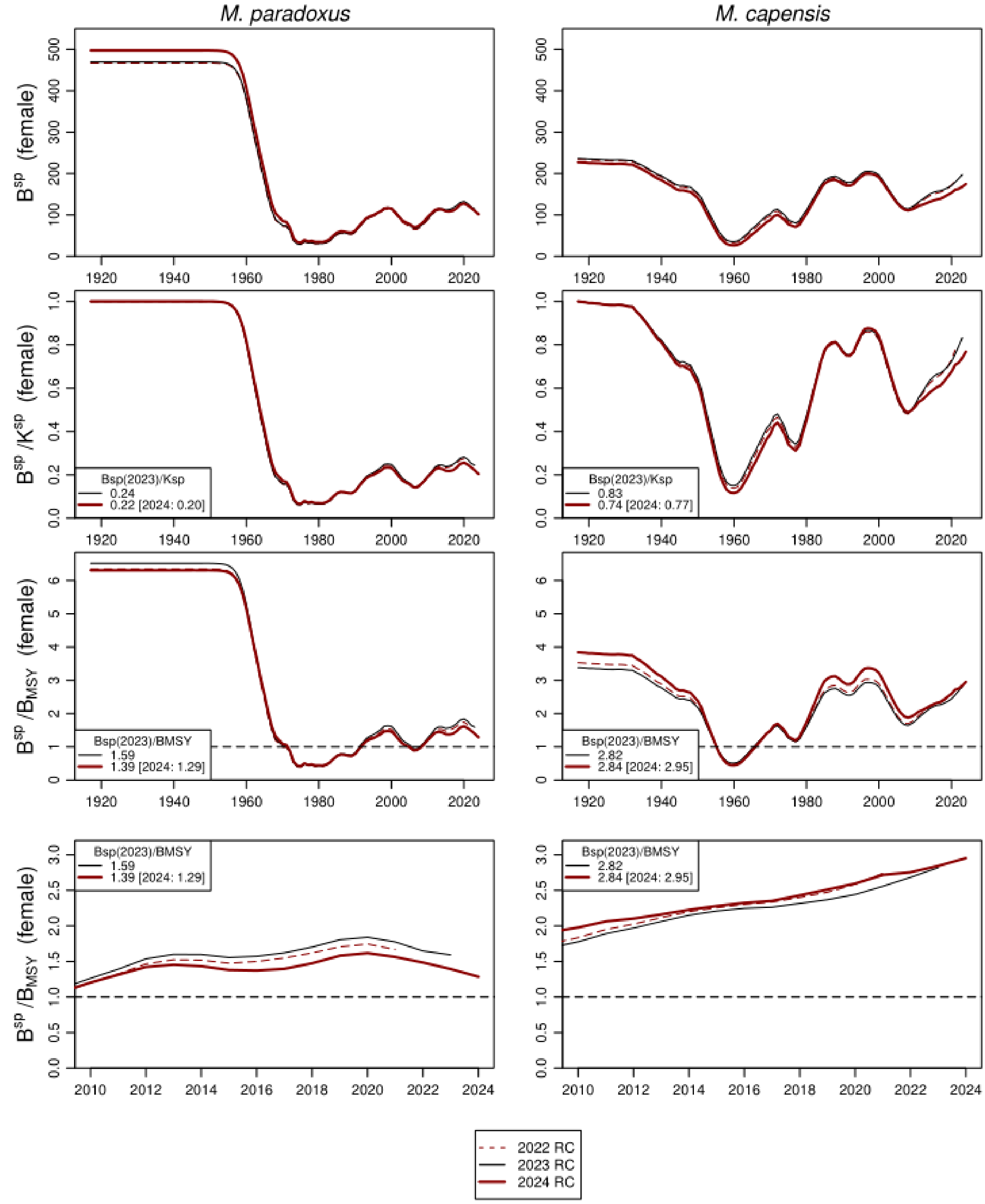


Figure 3: Female spawning biomass for the **2022, 2023 and 2024 RC OM**. Biomass is shown for each species in absolute terms (top row), relative to pristine spawning biomass (second row), relative to B_{MSY} (third row) and relative to B_{MSY} but for the 2010-2024 time period only (fourth row) and. The depletion levels have been listed in the figure legends for the 2023 and 2024 OMs for the year 2022, with the 2023 values for Run3 included in square brackets.

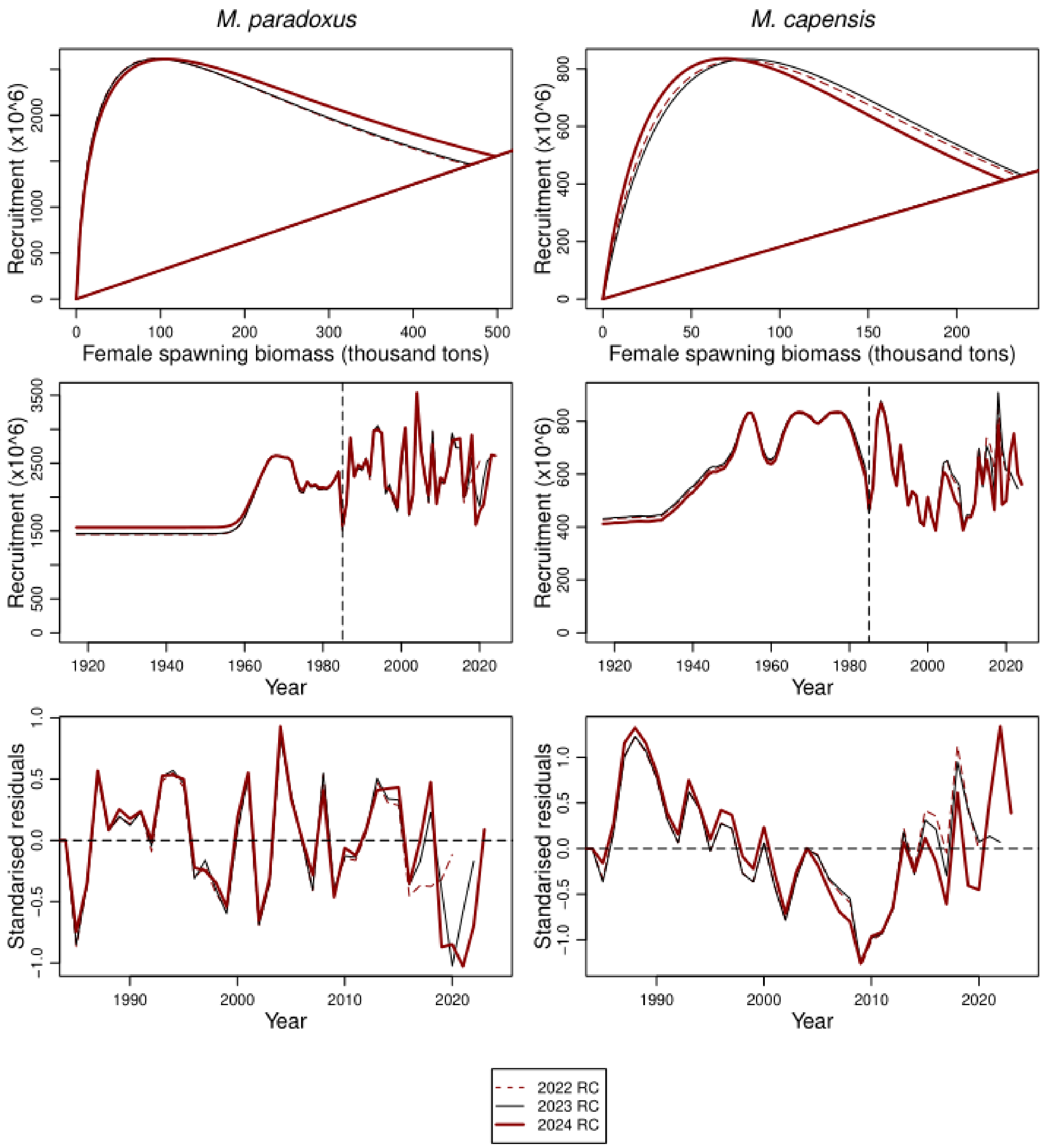


Figure 4: Further results for the **2022, 2023 and 2024 Ricker OMs.** The top row shows the estimated recruitment-spawning biomass relationship for the three assessment models, while the second row shows the estimated recruitment values and the bottom row the standardized estimated recruitment residuals.

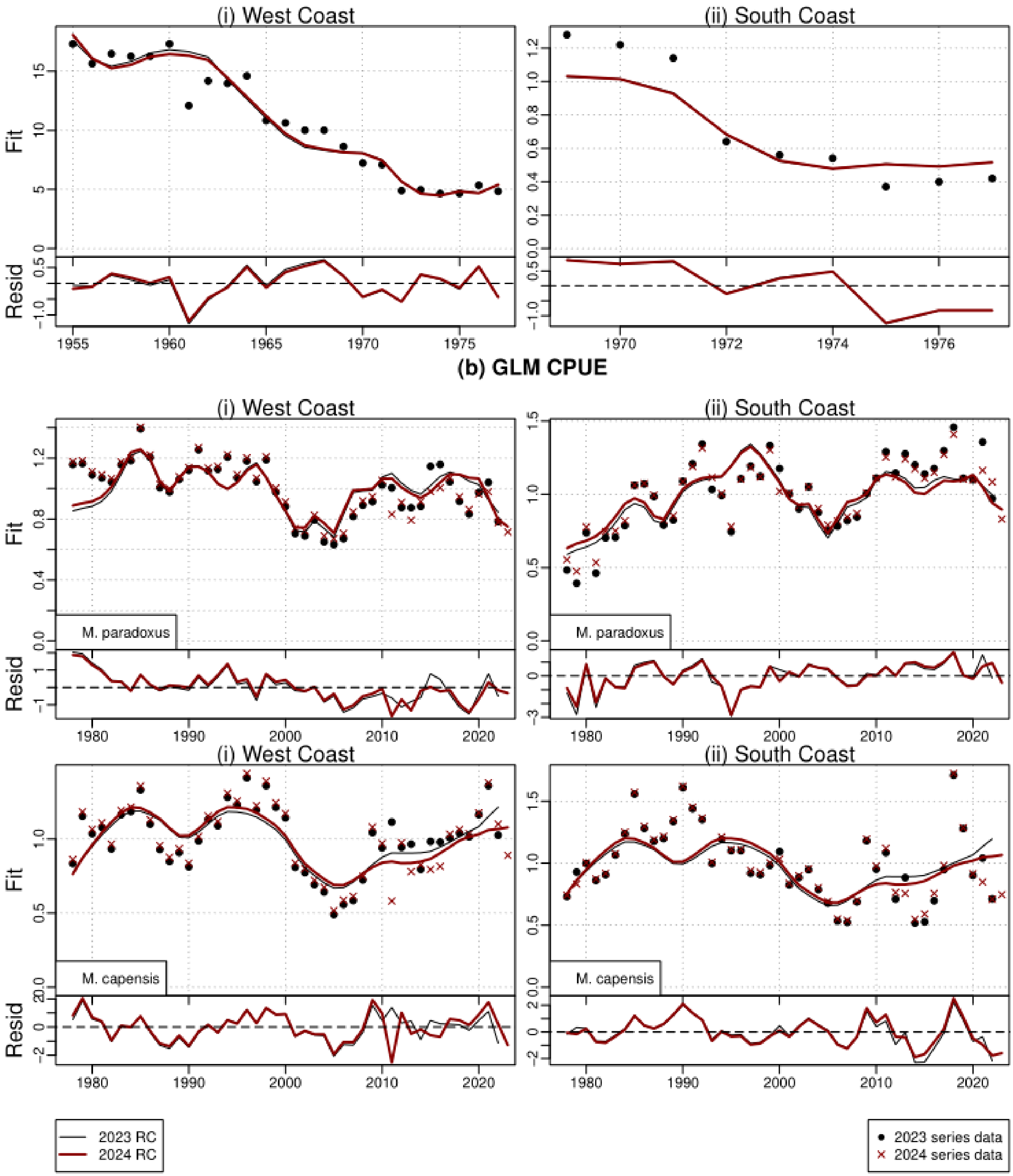


Figure 5: CPUE fits for the **2022, 2023 and 2024 Ricker OMs.** Fits to (A) the historical ICSEAF CPUE data and (B) the commercial GLM-standardized CPUE data are shown. Standardised residuals are shown underneath each plot. The updated 2024 GLM CPUE series has been shown by the red crosses.

Appendix A: Multi-modal likelihood for Ricker models.

For the Ricker models, the likelihood surface seems to be multi-modal, with the B^{sp}/B_{MSY} value seemingly not very well estimated. The aim of the jittering exercise is to try to explore the likelihood surface as thoroughly as possible to try find all of these modes. For RS02 and RS03 the difference in results for different modes were minimal. For RS01 (Ricker with central year 1952), however, one such instance of a multi-mode was found where the *M. capensis* stock was estimated to be much more depleted than what is normally estimated. For this result, the stock recruitment parameters were "switched" for the *M. capensis*, i.e. with a low *h* and high γ instead of the other way round. Even though the negative log-likelihood for this alternative run was marginally better than the one reported on in the main document (-3613.43 vs -3612.43), it was not kept as the best model because the associated B_{Msy}/K^{sp} estimate was low (0.14). The RS01 run reported on in the main text estimated B_{Msy}/K^{sp} at 0.16, which is probably about the lowest one could allow this value to go.

Table A. 1 lists key parameter estimates and Table A. 2 lists the negative log-likelihood components. Figure A. 1 plots the spawning biomass trajectories, Figure A. 2 plots the yield curves and Figure A. 3 the fits to the commercial CPUE data.

Even though the alternative run has been excluded on grounds of the low B_{MSY}/K^{sp} estimate for *M. paradoxus*, that estimate is not much lower than for the run reported on in the main text and there is little ground to choose between the two. It seems that the data are not able to distinguish between these two possible realities and as such OMP projections will be applied to this alternative run to assess whether the OMP is able to deal with such a situation should it in fact be the reality.

Table A. 1: Key assessment output and parameter estimates for the two RS01 runs. The green corresponds to the run reported on in the main text and the light blue the alternative run with slightly better negative log-likelihood and very different estimated status for *M. capensis*, but a B_{MSY}/K^{sp} less than 0.15.

M. paradoxus

−lnL	−3612.42	−3613.43
K^{sp} (f)	510	528
K^{sp} (m)	429	439
B_{MSY}	80	75
MSY	130	130
$B^{sp}{}_{2024}$	101	93
$B^{tot}{}_{2024}$	176	164
$B^{sp}{}_{2024}/K^{sp}$	0.20	0.18
$B^{sp}{}_{2024}/B_{MSY}$	1.25	1.25
B_{MSY}/K^{sp}	0.16	0.14

M. capensis

K^{sp} (f)	342	258
K^{sp} (m)	370	285
B_{MSY}	76	110
MSY	109	60
$B^{sp}{}_{2024}$	279	63
$B^{tot}{}_{2024}$	610	159
$B^{sp}{}_{2024}/K^{sp}$	0.82	0.25
$B^{sp}{}_{2024}/B_{MSY}$	3.69	0.58
B_{MSY}/K^{sp}	0.22	0.43

			M. paradoxus				M. capensis			
ln(K)	(3.50,10.00)		6.24	6.27			5.83	5.55		
h	(0.50,2.00)		1.63	1.57			1.74	0.93		
gamma	(0.00,2.00)		0.35	0.30			0.53	1.37		
Zeta	(−5.00,5.00)	1985	−0.33	−0.36			−0.10	−0.19		
		1986	−0.16	−0.16			0.08	−0.01		
		1987	0.26	0.26			0.48	0.29		
		1988	0.04	0.03			0.58	0.28		
		1989	0.12	0.11			0.52	0.24		
		1990	0.08	0.08			0.38	0.16		
		1991	0.11	0.11			0.18	−0.00		
		1992	0.00	−0.01			0.07	−0.12		
		1993	0.24	0.24			0.32	0.07		
		1994	0.24	0.24			0.16	−0.14		
		1995	0.22	0.23			−0.02	−0.30		
		1996	−0.10	−0.10			0.09	−0.18		
		1997	−0.11	−0.11			0.06	−0.15		
		1998	−0.16	−0.15			−0.15	−0.33		
		1999	−0.25	−0.24			−0.20	−0.38		
		2000	0.08	0.08			0.02	−0.16		
		2001	0.25	0.26			−0.18	−0.27		
		2002	−0.29	−0.29			−0.34	−0.36		
		2003	−0.14	−0.13			−0.11	−0.04		
		2004	0.42	0.43			0.02	0.23		
		2005	0.16	0.16			−0.05	0.30		
		2006	0.01	0.01			−0.17	0.27		
		2007	−0.13	−0.12			−0.26	0.19		
		2008	0.19	0.19			−0.30	0.15		
		2009	−0.21	−0.21			−0.50	−0.16		
		2010	−0.03	−0.02			−0.36	−0.12		
		2011	−0.06	−0.06			−0.34	−0.16		
		2012	0.04	0.04			−0.24	−0.03		
		2013	0.18	0.19			0.09	0.31		
		2014	0.19	0.19			−0.07	0.11		
		2015	0.19	0.20			0.10	0.26		
		2016	−0.16	−0.16			−0.00	0.09		
		2017	−0.00	−0.00			−0.24	−0.14		
		2018	0.21	0.22			0.33	0.31		
		2019	−0.39	−0.39			−0.14	−0.21		
		2020	−0.31	−0.31			−0.15	−0.20		
		2021	−0.29	−0.29			0.15	0.12		
		2022	−0.13	−0.14			0.25	0.24		
		2023	0.01	0.01			0.04	0.04		
sig_add	(0.00,0.50)		0.17	0.17			0.14	0.13		
sig_ICSEAF	(−20.0,20.0)	0.25								
sig_ICSEAF	(−20.0,20.0)	0.25								
sig_GLM	(−20.0,20.0)		0.15	0.15			0.15	0.15		
sig_GLM	(−20.0,20.0)		0.15	0.15			0.23	0.25		
qWCz1	(−20.0,20.0)						8.25	0.38		
qWCz2	(−20.0,20.0)						9.62	0.01		
qWCpar	(−20.0,20.0)		0.03	0.03						
r	(−20.0,20.0)		0.08	0.12						
gamma	(−20.0,20.0)						0.00	0.89		
			Old gear		New gear		Old gear		New gear	
exp(lnq WC sum)	(0.01,7.39)		1.76	1.78	1.55	1.57	0.80	1.44	0.55	1.00
exp(lnq WC win)	(0.01,7.39)		1.35	1.37			0.69	1.14		
exp(lnq SC spr)	(0.01,7.39)		0.87	0.88	0.80	0.81	0.66	1.72	0.45	1.16
exp(lnq SC aut)	(0.01,7.39)		1.14	1.15	0.97	0.98	0.81	1.99	0.54	1.33
			Males		Females		Males		Females	
AL theta0	(−20.0,20.0)		9.86	11.0	9.83	3.15	2.55	2.57	2.73	2.75
theta1	(0.15,1.00)		4.09	4.11	4.29	4.30	4.18	4.13	4.68	4.57
theta14	(0.00,10.00)		8.65	8.84	11.9	11.8	8.33	8.44	7.91	8.14
L2	(0.01,20.00)		24.6	24.5	22.0	21.9	26.7	26.5	25.5	25.3
L4	(0.01,20.00)		41.9	41.8	41.5	41.4	44.8	44.4	44.6	44.1
X	(0.01,20.00)		0.50	0.50			0.50	0.50		

Table A. 2: Negative log-likelihood components for the two runs. Table (a) shows the likelihoods in absolute terms, while the second row in table (b) shows the difference between the two runs.

(a) Negative log−likelihood components for various runs.

Run	GLM CPUE	ICSEAF CPUE	Survey abun.	Comm. CAL	Survey CAL	Recruit. resid.	ALKs	Penalties	Total (w/o pen.)
RS01 (main text)	−252.14	−38.07	−40.19	−1615.60	−1806.82	11.39	129.00	0.01	−3612.43
RS01 (alternative)	−249.87	−39.98	−41.92	−1616.21	−1803.29	9.75	128.07	0.01	−3613.44

(b) Negative log−likelihood components relative to the first row (grey highlight indicates a worse nll).

Run	GLM CPUE	ICSEAF CPUE	Survey abun.	Comm. CAL	Survey CAL	Recruit. resid.	ALKs	Penalties	Total (w/o pen.)
RS01 (main text)	−252.14	−38.07	−40.19	−1615.60	−1806.82	11.39	129.00	0.01	−3612.43
RS01 (alternative)	2.27	−1.91	−1.73	−0.62	3.53	−1.64	−0.92	0.00	−1.01

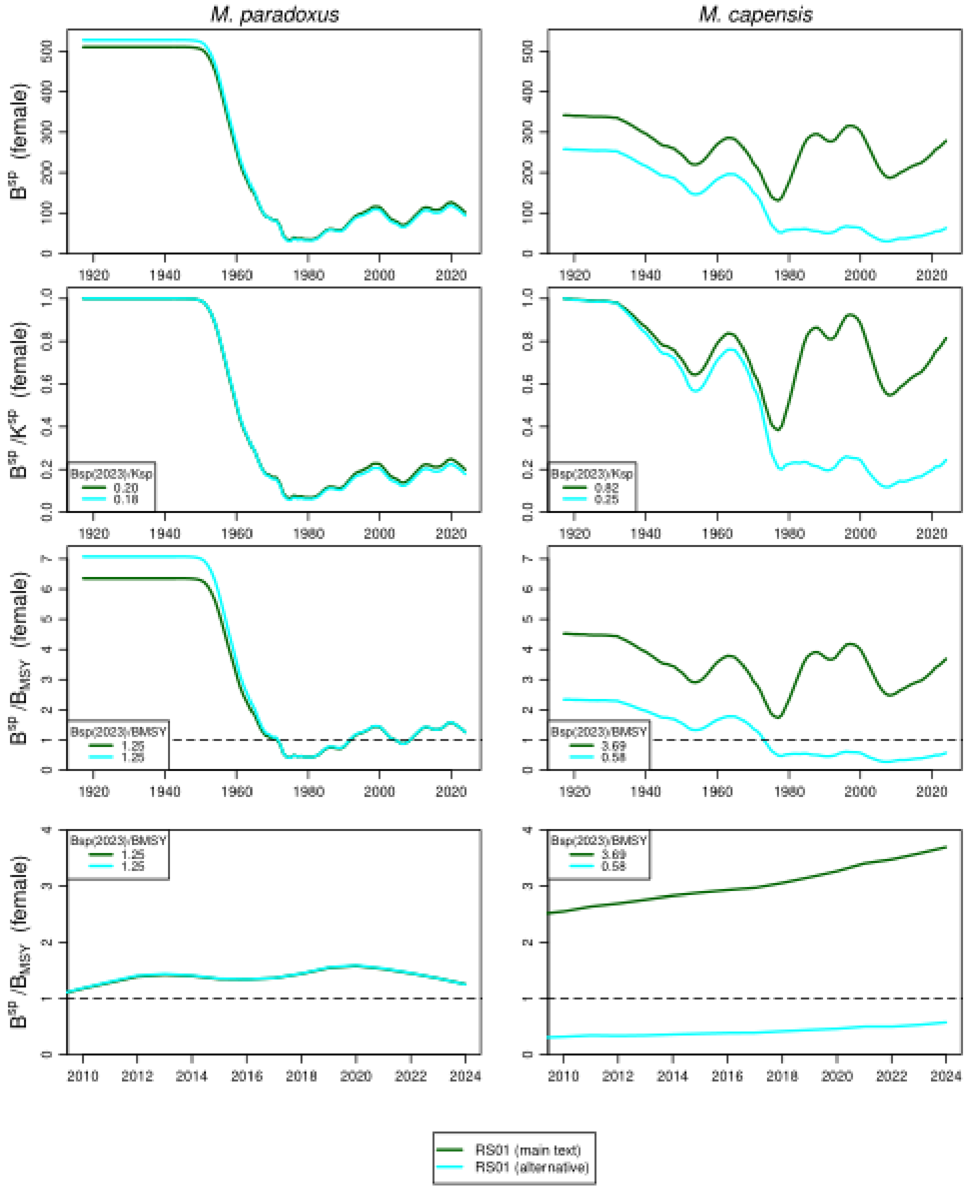


Figure A. 1: Female spawning biomass trajectories for the two RS01 runs. The top row shows spawning biomass in absolute terms, the second row relative to pristine biomass, the third row relative to B_{MSY} and the fourth row relative to B_{MSY} but restricted to the years 2010-2024.

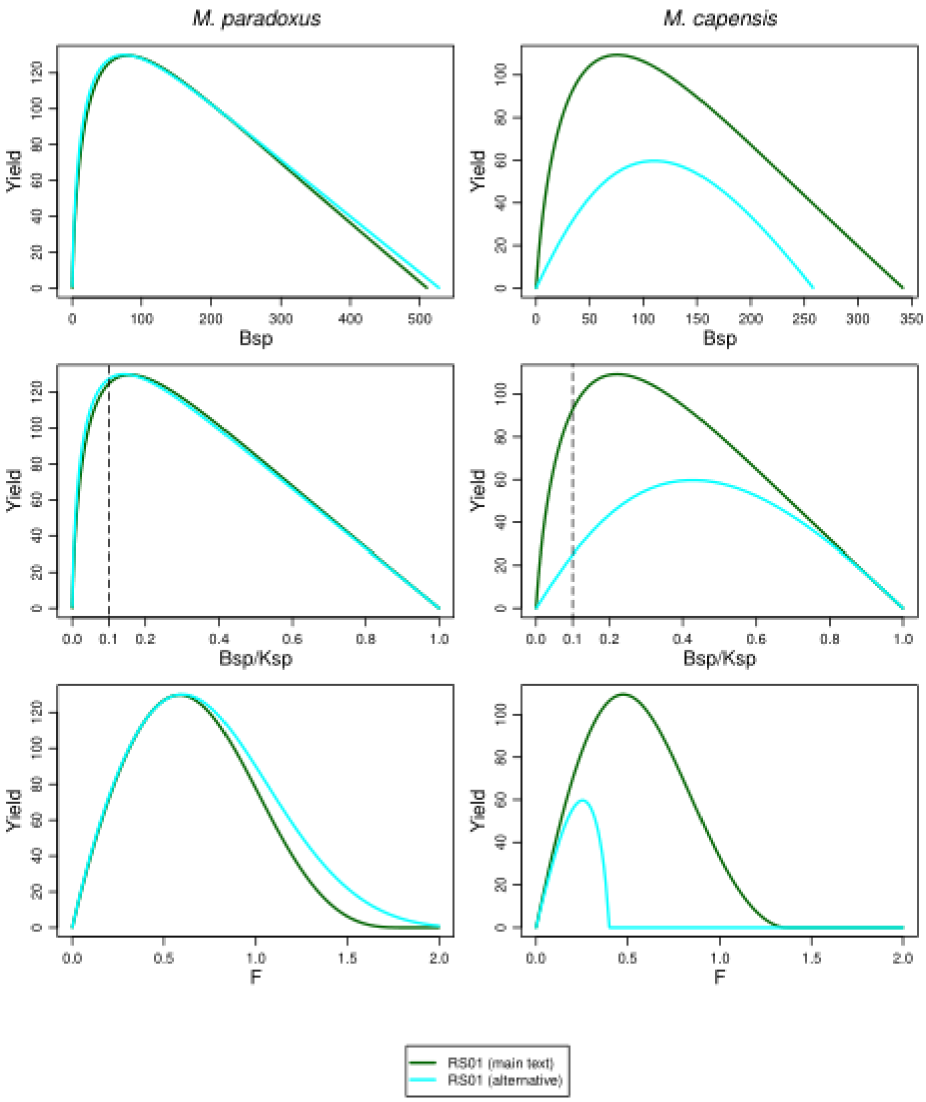


Figure A. 2: Plots of the yield curves for the two runs, showing a much reduced capacity for fishing pressure.

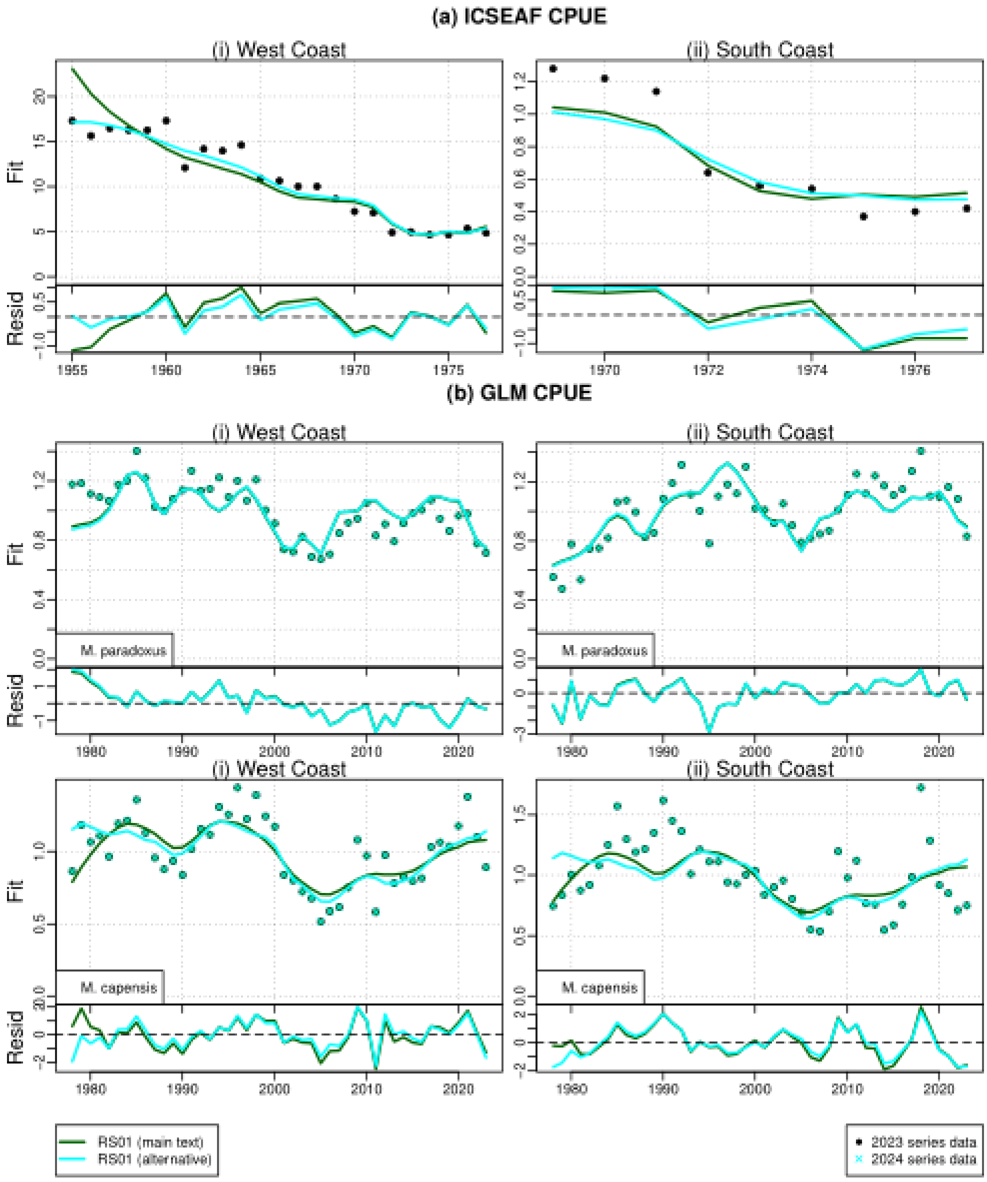


Figure A. 3: Fits to the abundance data for the two runs.

Appendix B: Detailed assessment results for the nine Reference Set Operating Models

This Appendix contains the detailed assessment results for the nine RS OMs. Each figure contains three panels, the first with plots for the three Ricker models, the second with plots for the Beverton-Holt models with *h*=0.90 and the third for Beverton-Holt with *h*=0.70.

The table below provides a summary of the plots that have been included.

Reference	**Description**
Table B. 1a-c	Parameter estimates for the nine RS OMs
Figure B. 4	Yield curves
Figure B. 5	Fits to survey relative abundance indices
Figure B. 6	Fits to commercial and survey catch-at-length data
Figure B. 7a	Commercial selectivity-at-length functions
Figure B. 7b	Commercial selectivity-at-age functions
Figure B. 8a	Survey selectivity-at-length functions
Figure B. 8b	Survey selectivity-at-age functions
Figure B. 9	Estimated growth curves

Table B. 1a: Parameter estimates for the **three Ricker models**: green for the model with central year 1952, red for the model with central year 1958 and yellow for the model with central year 1963. The first block reports the estimates for pristine spawning biomass, the stock recruitment *h* parameter and the Ricker γ parameter. The second block ("Zeta") reports estimates for the stock-recruitment residuals. In the third block the five parameters starting with "sig" reports the standard deviations for the CPUE series residuals, while the fourth block reports the values relating to the treatment of the historical ICSEAF CPUE data. The second to last block reports the estimates of the survey catchability coefficients (normal space, i.e. not logged), and the last block lists the estimates relating to the growth curves.

	M. paradoxus				*M. capensis*		
-lnL	-3612.42	-3613.91	-3612.80				
K^{sp} (f)	510	497	418	K^{sp} (f)	342	227	386
K^{sp} (m)	429	414	340	K^{sp} (m)	370	245	416
B_{MSY}	80	79	87	B_{MSY}	76	59	117
MSY	130	130	131	MSY	109	84	103
B^{sp}_{2024}	101	101	118	B^{sp}_{2024}	279	175	327
B^{tot}_{2024}	176	176	200	B^{tot}_{2024}	610	390	709
B^{sp}_{2024}/K^{sp}	0.20	0.20	0.28	B^{sp}_{2024}/K^{sp}	0.82	0.77	0.85
B^{sp}_{2024}/B_{MSY}	1.25	1.29	1.36	B^{sp}_{2024}/B_{MSY}	3.69	2.95	2.80
B_{MSY}/K^{sp}	0.16	0.16	0.21	B_{MSY}/K^{sp}	0.22	0.26	0.30

Parameters	Bounds		*M. paradoxus*						*M. capensis*					
ln(K)	(3.50,10.00)		6.24	6.21	6.03				5.83	5.43	5.95			
h	(0.50,2.00)		1.63	1.68	1.99				1.74	1.92	1.40			
gamma	(0.00,2.00)		0.35	0.36	0.58				0.53	0.73	0.79			
Zeta	(-5.00,5.00)	1985	-0.33	-0.34	-0.34				-0.10	-0.07	-0.10			
		1986	-0.16	-0.16	-0.15				0.08	0.12	0.07			
		1987	0.26	0.26	0.24				0.48	0.52	0.47			
		1988	0.04	0.04	0.05				0.58	0.60	0.57			
		1989	0.12	0.11	0.11				0.52	0.52	0.52			
		1990	0.08	0.08	0.08				0.38	0.38	0.38			
		1991	0.11	0.11	0.10				0.18	0.17	0.18			
		1992	0.00	-0.00	0.01				0.07	0.07	0.06			
		1993	0.24	0.24	0.23				0.32	0.34	0.32			
		1994	0.24	0.24	0.24				0.16	0.20	0.16			
		1995	0.22	0.22	0.20				-0.02	0.04	-0.03			
		1996	-0.10	-0.10	-0.08				0.09	0.19	0.09			
		1997	-0.11	-0.11	-0.12				0.06	0.17	0.05			
		1998	-0.16	-0.15	-0.15				-0.15	-0.04	-0.16			
		1999	-0.25	-0.24	-0.24				-0.20	-0.10	-0.21			
		2000	0.08	0.08	0.08				0.02	0.11	0.01			
		2001	0.25	0.25	0.24				-0.18	-0.12	-0.19			
		2002	-0.29	-0.29	-0.29				-0.34	-0.32	-0.35			
		2003	-0.14	-0.14	-0.16				-0.11	-0.12	-0.12			
		2004	0.42	0.42	0.41				0.02	0.00	0.01			
		2005	0.16	0.15	0.14				-0.05	-0.08	-0.05			
		2006	0.01	0.01	0.01				-0.17	-0.20	-0.16			
		2007	-0.13	-0.13	-0.15				-0.26	-0.31	-0.25			
		2008	0.19	0.18	0.19				-0.30	-0.36	-0.28			
		2009	-0.21	-0.21	-0.21				-0.50	-0.57	-0.49			
		2010	-0.03	-0.03	-0.04				-0.36	-0.43	-0.35			
		2011	-0.06	-0.06	-0.05				-0.34	-0.41	-0.33			
		2012	0.04	0.04	0.03				-0.24	-0.29	-0.23			
		2013	0.18	0.18	0.20				0.09	0.03	0.10			
		2014	0.19	0.19	0.20				-0.07	-0.11	-0.07			
		2015	0.19	0.19	0.21				0.10	0.05	0.10			
		2016	-0.16	-0.16	-0.15				-0.00	-0.06	0.01			
		2017	-0.00	0.00	0.02				-0.24	-0.28	-0.23			
		2018	0.21	0.21	0.21				0.33	0.28	0.33			
		2019	-0.39	-0.39	-0.37				-0.14	-0.18	-0.14			
		2020	-0.31	-0.31	-0.29				-0.15	-0.16	-0.14			
		2021	-0.29	-0.28	-0.25				0.15	0.15	0.15			
		2022	-0.13	-0.13	-0.12				0.25	0.25	0.25			
		2023	0.01	0.01	0.01				0.04	0.04	0.04			
sig_add	(0.00,0.50)		0.17	0.17	0.17				0.14	0.14	0.14			
sig_ICSEAF	(-20.0,20.0)	0.25												
sig_ICSEAF	(-20.0,20.0)	0.25												
sig_GLM	(-20.0,20.0)		0.15	0.15	0.15				0.15	0.15	0.15			
sig_GLM	(-20.0,20.0)		0.15	0.15	0.15				0.23	0.22	0.23			
qWCz1	(-20.0,20.0)								8.25	0.45	0.54			
qWCz2	(-20.0,20.0)								9.62	0.06	0.04			
qWCpar	(-20.0,20.0)		0.03	0.03	0.03									
r	(-20.0,20.0)		0.08	0.08	0.09									
gamma	(-20.0,20.0)								0.00	0.67	0.88			
			Old gear			New gear			Old gear			New gear		
exp(lnq WC sum)	(0.01,7.39)		1.76	1.76	1.67	1.55	1.55	1.47	0.80	1.05	0.73	0.55	0.72	0.50
exp(lnq WC win)	(0.01,7.39)		1.35	1.35	1.29				0.69	0.90	0.62			
exp(lnq SC spr)	(0.01,7.39)		0.87	0.88	0.84	0.80	0.80	0.76	0.66	0.92	0.59	0.45	0.62	0.40
exp(lnq SC aut)	(0.01,7.39)		1.14	1.14	1.09	0.97	0.97	0.93	0.81	1.11	0.72	0.54	0.75	0.48
			Males			Females			Males			Females		
AL theta0	(-20.0,20.0)		9.86	10.3	9.96	9.83	10.3	17.2	2.55	2.55	2.49	2.73	2.73	2.75
theta1	(0.15,1.00)		4.09	4.09	4.12	4.29	4.31	4.33	4.18	4.14	4.18	4.68	4.65	4.69
theta14	(0.00,10.00)		8.65	8.66	7.96	11.9	11.9	11.7	8.33	8.52	8.29	7.91	7.98	7.88
L2	(0.01,20.00)		24.6	24.5	24.8	22.0	22.0	22.3	26.7	26.6	26.8	25.5	25.4	25.5
L4	(0.01,20.00)		41.9	41.8	41.7	41.5	41.5	41.7	44.8	44.7	44.8	44.6	44.5	44.6
X	(0.01,20.00)		0.50	0.50	0.50				0.50	0.50	0.50			

Table B. 1b: Parameter estimates for the three **Beverton-Holt models with *h*=0.90**.

	M. paradoxus			*M. capensis*		
-lnL	-3584.67	-3576.45	-3573.84			
K^{sp} (f)	981	978	959	1090	1062	1185
K^{sp} (m)	744	822	826	1178	1163	1280
B_{MSY}	140	136	129	168	164	182
MSY	117	118	121	162	157	176
B^{sp}_{2024}	177	171	201	926	897	1021
B^{tot}_{2024}	273	279	334	1973	1925	2172
B^{sp}_{2024}/K^{sp}	0.18	0.18	0.21	0.85	0.85	0.86
B^{sp}_{2024}/B_{MSY}	1.26	1.26	1.56	5.52	5.48	5.60
B_{MSY}/K^{sp}	0.14	0.14	0.13	0.15	0.15	0.15

Parameters	Bounds		*M. paradoxus*						*M. capensis*					
ln(K)	(3.50,10.00)		6.89	6.89	6.87				6.99	6.97	7.08			
h	(0.50,2.00)		0.90	0.90	0.90				0.90	0.90	0.90			
gamma	(0.00,2.00)													
Zeta	(-5.00,5.00)	1985	-0.37	-0.43	-0.39				-0.12	-0.12	-0.12			
		1986	-0.10	-0.11	-0.09				-0.01	0.01	-0.01			
		1987	0.42	0.39	0.34				0.33	0.34	0.33			
		1988	0.05	0.02	0.06				0.43	0.42	0.42			
		1989	0.17	0.12	0.10				0.40	0.40	0.40			
		1990	0.10	0.07	0.08				0.27	0.29	0.27			
		1991	0.24	0.16	0.11				0.08	0.08	0.08			
		1992	-0.05	0.00	0.06				-0.03	-0.03	-0.04			
		1993	0.46	0.38	0.31				0.18	0.18	0.18			
		1994	0.13	0.19	0.24				0.01	-0.00	0.01			
		1995	0.41	0.30	0.19				-0.21	-0.21	-0.21			
		1996	-0.20	-0.19	-0.15				-0.15	-0.16	-0.16			
		1997	-0.10	-0.13	-0.18				-0.21	-0.21	-0.22			
		1998	-0.05	-0.15	-0.17				-0.40	-0.41	-0.41			
		1999	-0.48	-0.38	-0.33				-0.42	-0.42	-0.42			
		2000	0.16	0.11	0.05				-0.19	-0.19	-0.19			
		2001	0.36	0.31	0.25				-0.33	-0.32	-0.33			
		2002	-0.47	-0.39	-0.33				-0.41	-0.41	-0.41			
		2003	-0.19	-0.14	-0.15				-0.13	-0.14	-0.13			
		2004	0.63	0.54	0.51				0.04	0.05	0.04			
		2005	0.14	0.15	0.16				0.02	0.01	0.02			
		2006	0.19	0.08	0.08				-0.06	-0.04	-0.06			
		2007	-0.45	-0.29	-0.29				-0.08	-0.09	-0.08			
		2008	0.57	0.39	0.34				-0.10	-0.10	-0.10			
		2009	-0.63	-0.39	-0.31				-0.28	-0.29	-0.28			
		2010	0.19	0.09	0.00				-0.14	-0.14	-0.14			
		2011	-0.10	-0.09	-0.05				-0.13	-0.13	-0.12			
		2012	-0.04	0.02	-0.04				-0.06	-0.07	-0.06			
		2013	0.37	0.30	0.30				0.25	0.26	0.25			
		2014	0.06	0.14	0.14				0.07	0.07	0.07			
		2015	0.45	0.33	0.30				0.24	0.26	0.24			
		2016	-0.44	-0.23	-0.18				0.19	0.15	0.19			
		2017	0.10	0.08	0.10				-0.07	-0.07	-0.07			
		2018	0.42	0.38	0.28				0.47	0.48	0.47			
		2019	-0.91	-0.71	-0.53				0.03	0.01	0.04			
		2020	-0.31	-0.29	-0.30				-0.03	-0.02	-0.03			
		2021	-0.41	-0.35	-0.28				0.22	0.22	0.22			
		2022	-0.30	-0.25	-0.21				0.29	0.28	0.29			
		2023	-0.05	-0.03	-0.02				0.05	0.05	0.05			
sig_add	(0.00,0.50)		0.17	0.18	0.20				0.14	0.14	0.14			
sig_ICSEAF	(-20.0,20.0)	0.25												
sig_ICSEAF	(-20.0,20.0)	0.25												
sig_GLM	(-20.0,20.0)		0.15	0.15	0.15				0.16	0.16	0.16			
sig_GLM	(-20.0,20.0)		0.16	0.25	0.29				0.24	0.24	0.24			
qWCz1	(-20.0,20.0)								9.84	0.63	9.55			
qWCz2	(-20.0,20.0)								5.20	0.01	0.01			
qWCpar	(-20.0,20.0)		0.03	0.03	0.02									
r	(-20.0,20.0)		0.05	0.07	0.08									
gamma	(-20.0,20.0)								0.03	0.98	1.00			
			Old gear			New gear			Old gear			New gear		
exp(lnq WC sum)	(0.01,7.39)		1.54	1.56	1.41	1.37	1.39	1.25	0.33	0.33	0.30	0.22	0.23	0.20
exp(lnq WC win)	(0.01,7.39)		1.18	1.13	0.99				0.28	0.29	0.25			
exp(lnq SC spr)	(0.01,7.39)		0.77	0.78	0.60	0.70	0.71	0.55	0.25	0.26	0.23	0.17	0.17	0.15
exp(lnq SC aut)	(0.01,7.39)		1.00	1.01	0.71	0.85	0.87	0.61	0.30	0.32	0.28	0.20	0.21	0.18
			Males			Females			Males			Females		
AL theta0	(-20.0,20.0)		10.1	19.1	6.50	9.33	3.26	12.2	2.44	2.50	2.44	2.77	2.63	2.78
theta1	(0.15,1.00)		4.04	4.18	4.31	4.41	4.39	4.46	4.22	4.21	4.21	4.72	4.74	4.72
theta14	(0.00,10.00)		9.76	8.65	7.67	12.2	11.6	10.9	8.00	8.11	8.01	7.79	7.79	7.79
L2	(0.01,20.00)		24.8	24.8	25.2	22.3	22.2	23.0	26.9	26.6	26.9	25.6	25.6	25.7
L4	(0.01,20.00)		41.3	42.1	42.3	41.9	41.8	42.1	45.0	45.0	45.0	44.8	44.7	44.8
X	(0.01,20.00)		0.50	0.50	0.50				0.50	0.50	0.50			

Table B. 1c: Parameter estimates for the three **Beverton-Holt models with *h*=0.70**.

	M. paradoxus			*M. capensis*		
-lnL	-3579.41	-3568.33	-3572.49			
K^{sp} (f)	1404	1382	1482	411	1363	1685
K^{sp} (m)	1157	1141	1219	450	1471	1816
B_{MSY}	340	335	357	110	352	435
MSY	114	113	120	49	147	181
$B^{sp}{}_{2024}$	420	401	545	66	1166	1486
$B^{tot}{}_{2024}$	685	654	900	165	2474	3143
$B^{sp}{}_{2024}/K^{sp}$	0.30	0.29	0.37	0.16	0.86	0.88
$B^{sp}{}_{2024}/B_{MSY}$	1.24	1.20	1.53	0.60	3.31	3.42
B_{MSY}/K^{sp}	0.24	0.24	0.24	0.27	0.26	0.26

Parameters	Bounds		*M. paradoxus*						*M. capensis*					
ln(K)	(3.50,10.00)		7.25	7.23	7.30				6.02	7.22	7.43			
h	(0.50,2.00)		0.70	0.70	0.70				0.70	0.70	0.70			
gamma	(0.00,2.00)													
Zeta	(-5.00,5.00)	1985	-0.40	-0.40	-0.38				-0.05	-0.14	-0.14			
		1986	-0.09	-0.09	-0.07				0.08	-0.02	-0.03			
		1987	0.37	0.36	0.36				0.36	0.32	0.31			
		1988	0.07	0.06	0.08				0.34	0.42	0.42			
		1989	0.10	0.09	0.10				0.30	0.40	0.40			
		1990	0.09	0.08	0.09				0.20	0.26	0.26			
		1991	0.12	0.12	0.11				0.02	0.07	0.07			
		1992	0.06	0.05	0.07				-0.09	-0.05	-0.05			
		1993	0.32	0.32	0.32				0.11	0.17	0.17			
		1994	0.27	0.26	0.28				-0.08	-0.01	-0.01			
		1995	0.21	0.21	0.19				-0.24	-0.23	-0.23			
		1996	-0.12	-0.13	-0.11				-0.11	-0.18	-0.18			
		1997	-0.19	-0.19	-0.20				-0.09	-0.24	-0.24			
		1998	-0.20	-0.21	-0.22				-0.28	-0.43	-0.43			
		1999	-0.37	-0.37	-0.35				-0.33	-0.44	-0.44			
		2000	0.04	0.04	0.03				-0.11	-0.21	-0.21			
		2001	0.18	0.18	0.13				-0.25	-0.34	-0.35			
		2002	-0.38	-0.38	-0.37				-0.36	-0.42	-0.42			
		2003	-0.19	-0.18	-0.20				-0.07	-0.13	-0.13			
		2004	0.49	0.50	0.46				0.17	0.05	0.05			
		2005	0.14	0.15	0.14				0.21	0.03	0.03			
		2006	0.08	0.08	0.06				0.16	-0.04	-0.04			
		2007	-0.24	-0.24	-0.22				0.07	-0.06	-0.06			
		2008	0.37	0.37	0.33				0.02	-0.07	-0.07			
		2009	-0.25	-0.26	-0.22				-0.25	-0.26	-0.25			
		2010	0.02	0.03	-0.00				-0.19	-0.12	-0.11			
		2011	-0.05	-0.05	-0.05				-0.22	-0.10	-0.09			
		2012	-0.03	-0.02	-0.03				-0.09	-0.04	-0.04			
		2013	0.27	0.27	0.25				0.23	0.28	0.28			
		2014	0.16	0.17	0.18				0.06	0.09	0.09			
		2015	0.30	0.30	0.28				0.20	0.26	0.27			
		2016	-0.17	-0.17	-0.15				0.05	0.21	0.22			
		2017	0.13	0.13	0.14				-0.16	-0.06	-0.06			
		2018	0.28	0.28	0.24				0.29	0.48	0.49			
		2019	-0.55	-0.56	-0.51				-0.19	0.04	0.04			
		2020	-0.31	-0.31	-0.30				-0.17	-0.03	-0.03			
		2021	-0.28	-0.28	-0.24				0.15	0.22	0.21			
		2022	-0.21	-0.21	-0.18				0.26	0.28	0.28			
		2023	-0.02	-0.02	-0.01				0.04	0.05	0.05			
sig_add	(0.00,0.50)		0.19	0.19	0.20				0.13	0.14	0.14			
sig_ICSEAF	(-20.0,20.0)	0.25												
sig_ICSEAF	(-20.0,20.0)	0.25												
sig_GLM	(-20.0,20.0)		0.15	0.15	0.15				0.15	0.16	0.16			
sig_GLM	(-20.0,20.0)		0.27	0.29	0.29				0.25	0.24	0.24			
qWCz1	(-20.0,20.0)								1.36	0.01	0.01			
qWCz2	(-20.0,20.0)								0.01	0.09	0.05			
qWCpar	(-20.0,20.0)		0.02	0.02	0.01									
r	(-20.0,20.0)		0.21	0.07	0.07									
gamma	(-20.0,20.0)								0.99	0.06	0.09			
			Old gear			New gear			Old gear			New gear		
exp(lnq WC sum)	(0.01,7.39)		1.14	1.16	1.00	1.02	1.03	0.90	1.43	0.27	0.22	0.99	0.18	0.15
exp(lnq WC win)	(0.01,7.39)		0.77	0.78	0.66				1.10	0.23	0.18			
exp(lnq SC spr)	(0.01,7.39)		0.53	0.54	0.46	0.49	0.50	0.42	1.72	0.20	0.16	1.15	0.13	0.11
exp(lnq SC aut)	(0.01,7.39)		0.67	0.69	0.58	0.58	0.60	0.49	1.95	0.25	0.20	1.30	0.16	0.13
			Males			Females			Males			Females		
AL theta0	(-20.0,20.0)		9.20	7.81	8.73	10.6	10.3	9.20	2.41	2.43	2.38	2.79	2.77	2.78
theta1	(0.15,1.00)		4.36	4.34	4.43	4.48	4.50	4.54	4.09	4.21	4.21	4.58	4.73	4.73
theta14	(0.00,10.00)		6.95	7.28	6.50	11.2	11.2	11.0	8.64	8.02	8.00	8.12	7.79	7.79
L2	(0.01,20.00)		25.3	25.2	25.3	22.9	22.9	23.0	26.6	26.9	27.0	25.3	25.7	25.7
L4	(0.01,20.00)		42.5	42.4	42.5	42.4	42.4	42.5	44.4	45.0	45.0	44.1	44.8	44.8
X	(0.01,20.00)		0.50	0.50	0.50				0.50	0.50	0.50			

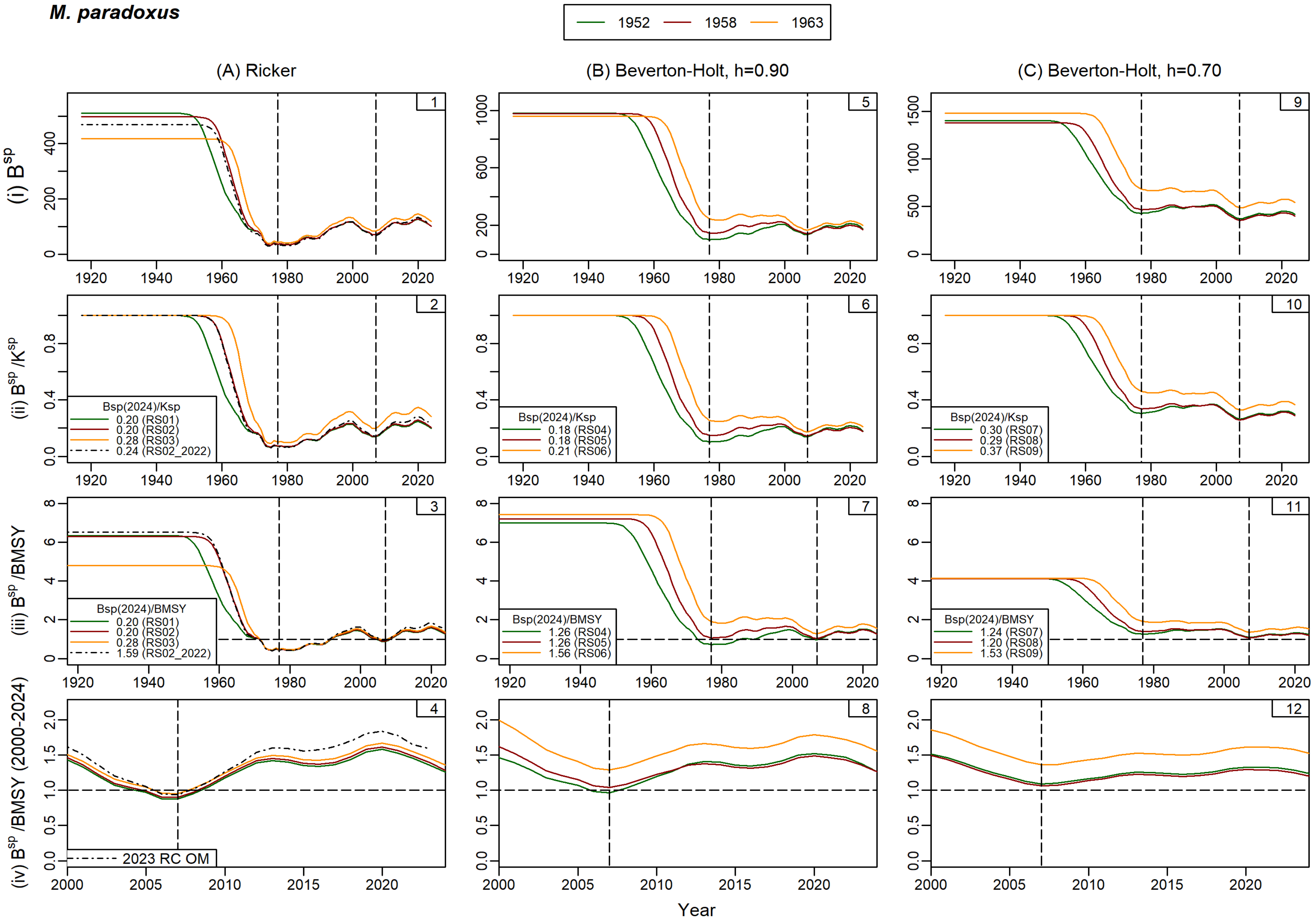
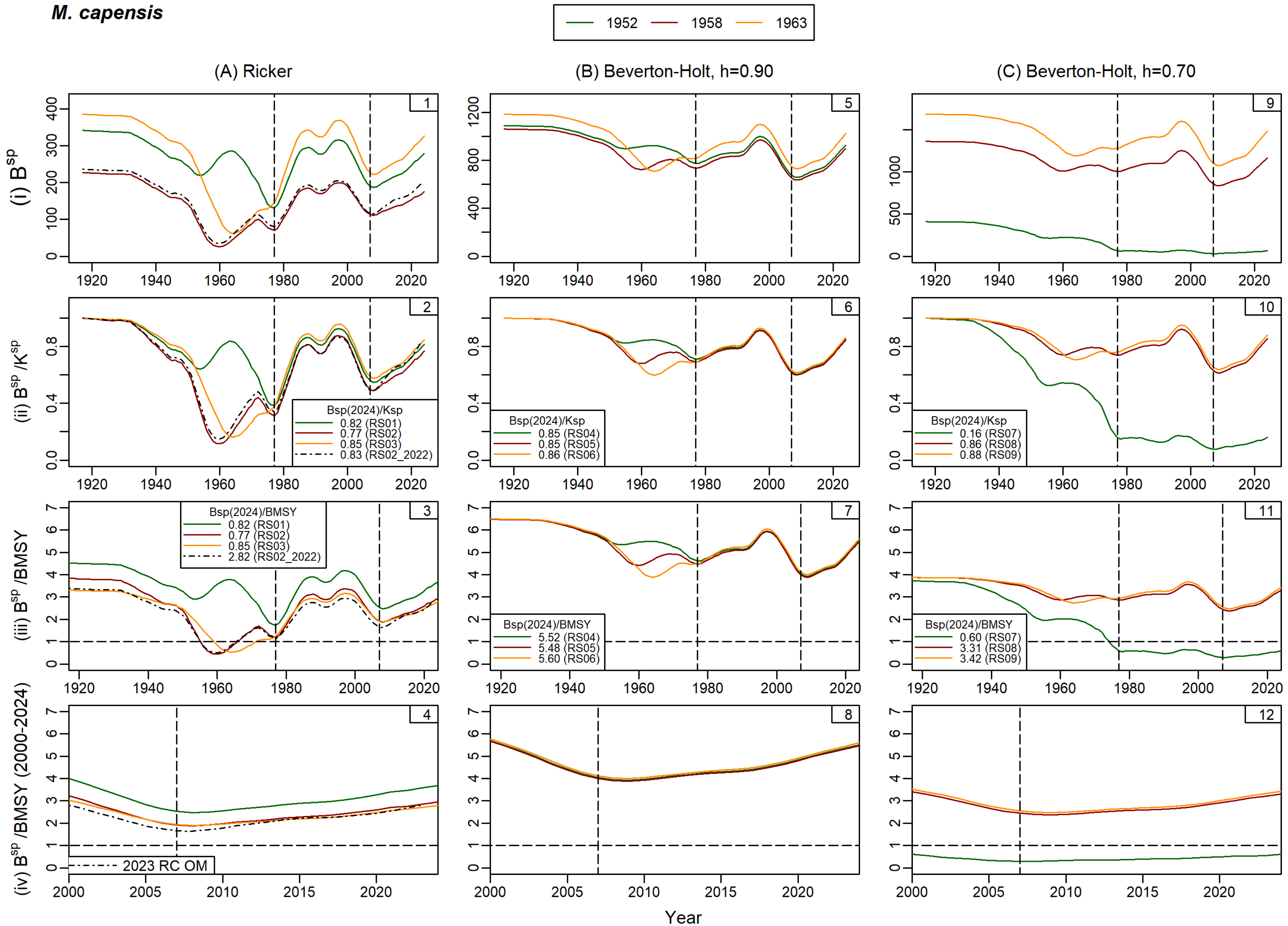


Figure B. 1a: Female spawning biomass trajectories are shown for ***M. paradoxus*** for smaller groupings of models. In the plots, yellow lines have been used for the models with the central year of shift occurring in 1952, blue lines for the 1958 models and red lines for the 1963 models. The corresponding Oct 2021 model has been included in the first column with the RS Ricker models (black dash-dot lines).



Error! Reference source not found.**b:** Female spawning biomass trajectories are shown for ***M. capensis*** for the smaller groupings of models (as in **Error! Reference source not found.**a).

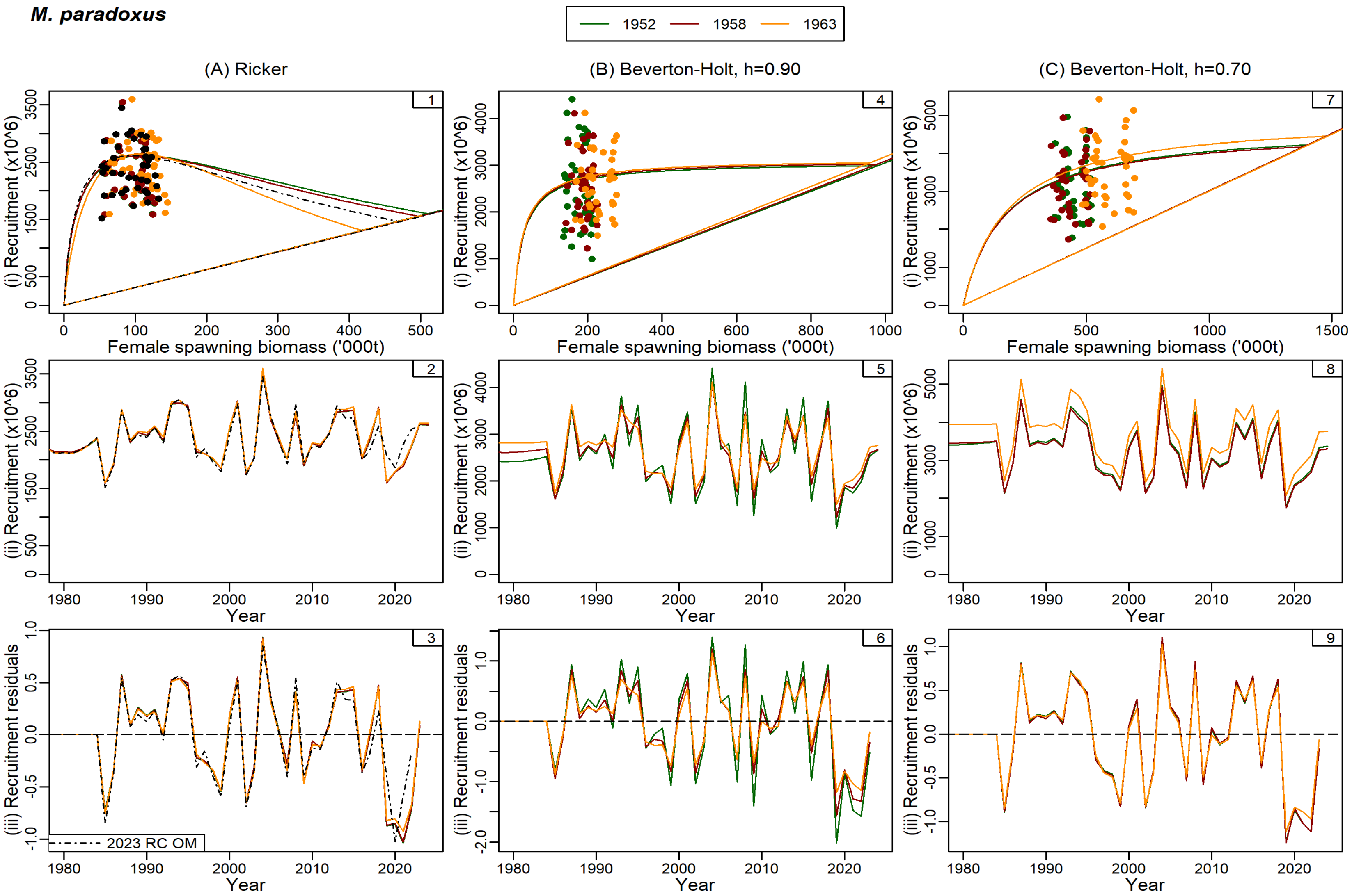
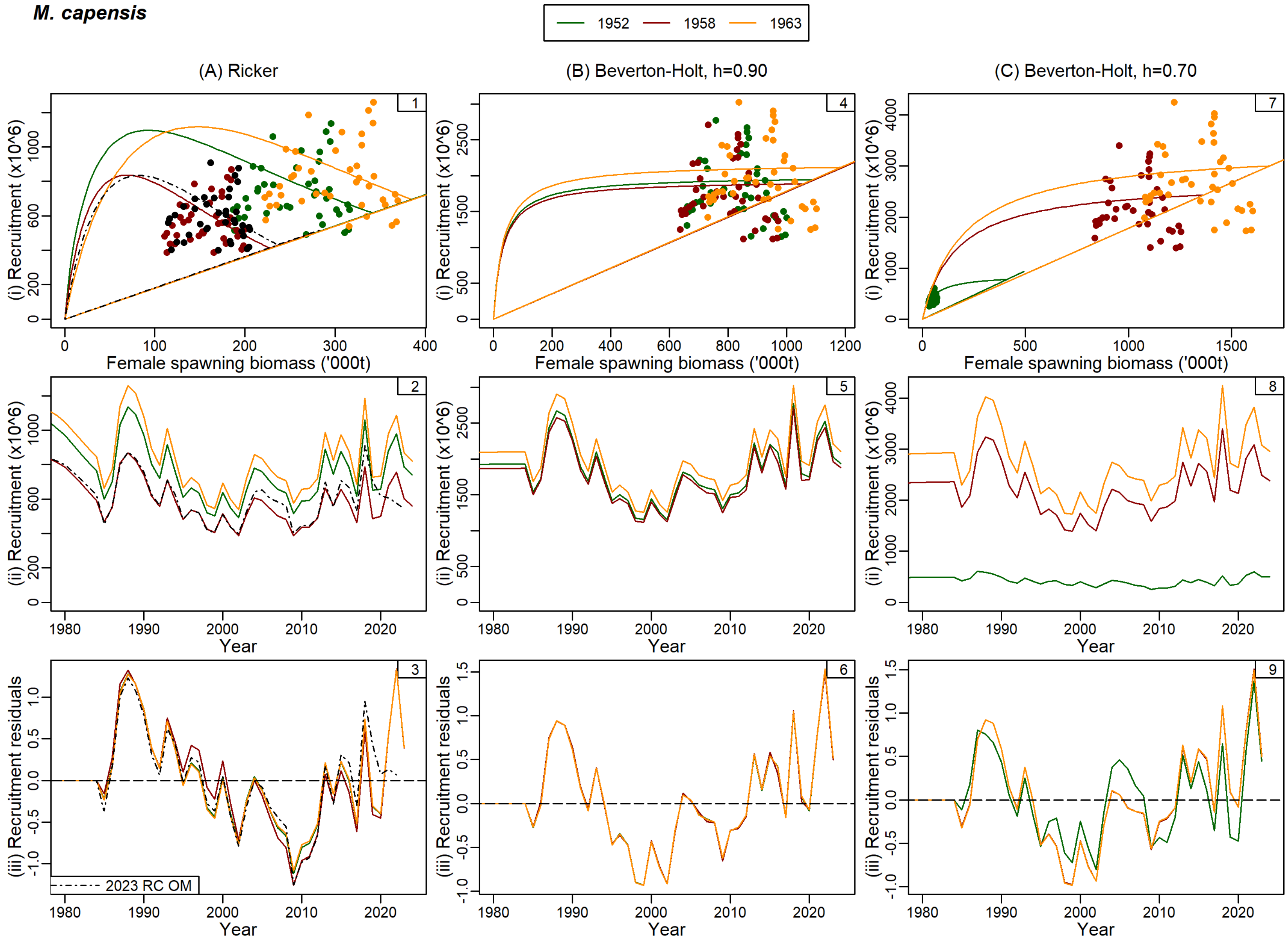


Figure B. 2: a: Stock-recruitment plots together with recruitment time series and residuals about the stock-recruitment curves are shown for ***M. paradoxus*** for the smaller groupings of models. In the interest of clarity, the "data" are shown for a selection of models only. The straight lines through the origin in the stock-recruitment plots are replacement lines.



Error! Reference source not found.**b:** Stock recruitment plots are shown for ***M. capensis*** for the smaller groupings of models (as in **Error! Reference source not found.**a).

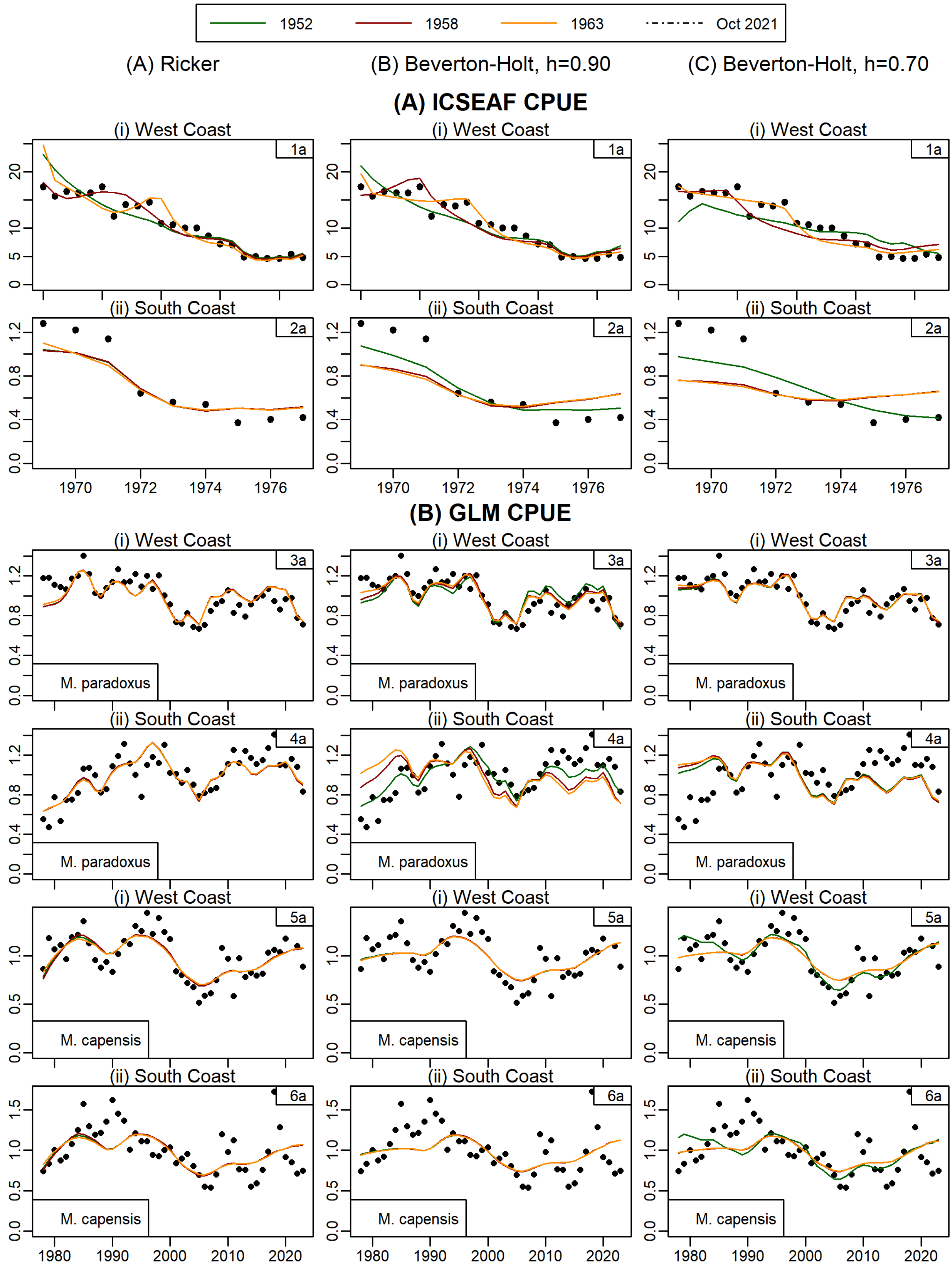


Figure B. 3:: Fits to the ICSEAF and commercial CPUE data.

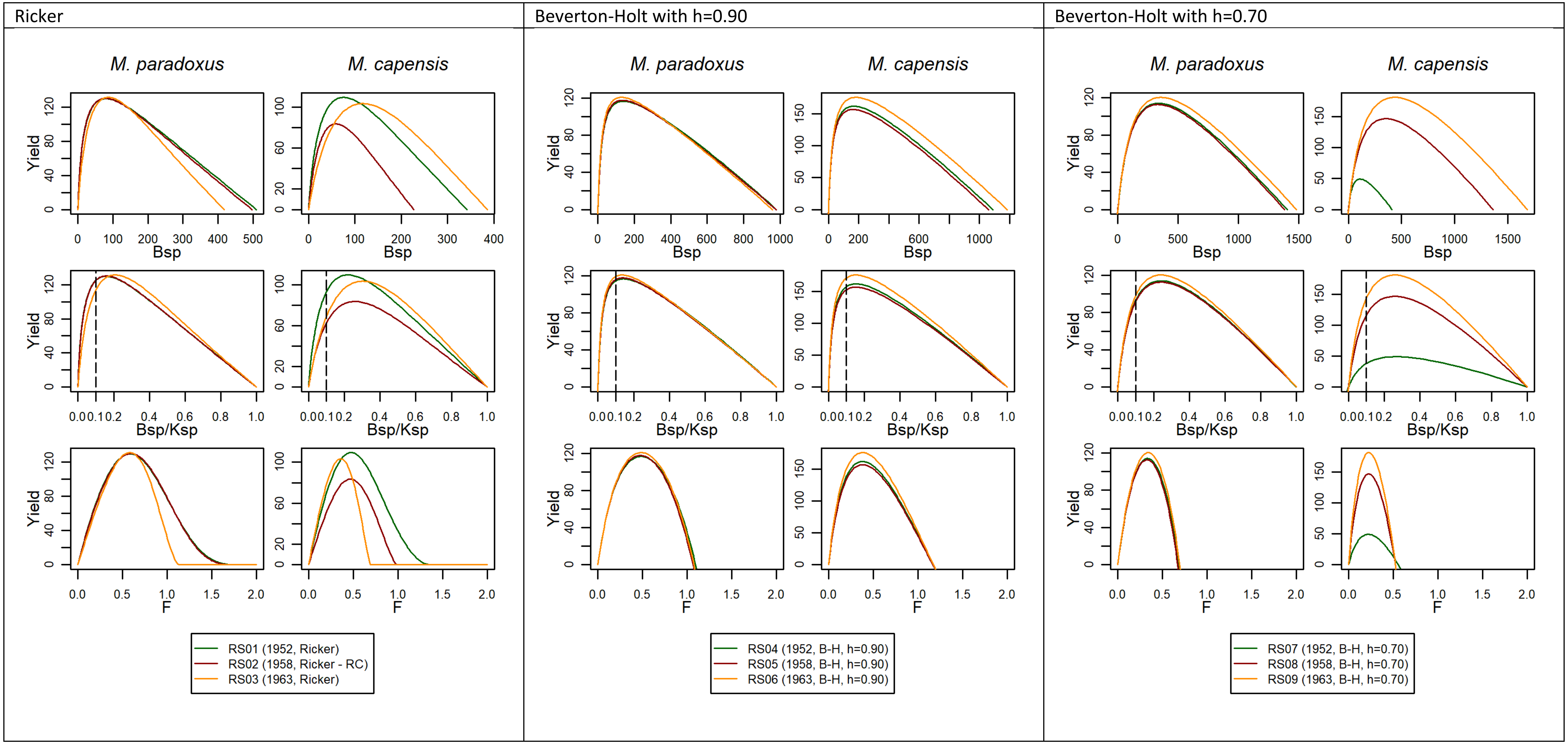


Figure B. 4: The estimated yield curve (yield in thousand tons) is plotted against female spawning biomass (top panel), female spawning biomass relative to pristine levels (middle panel) and fishing mortality rate F (bottom panel).

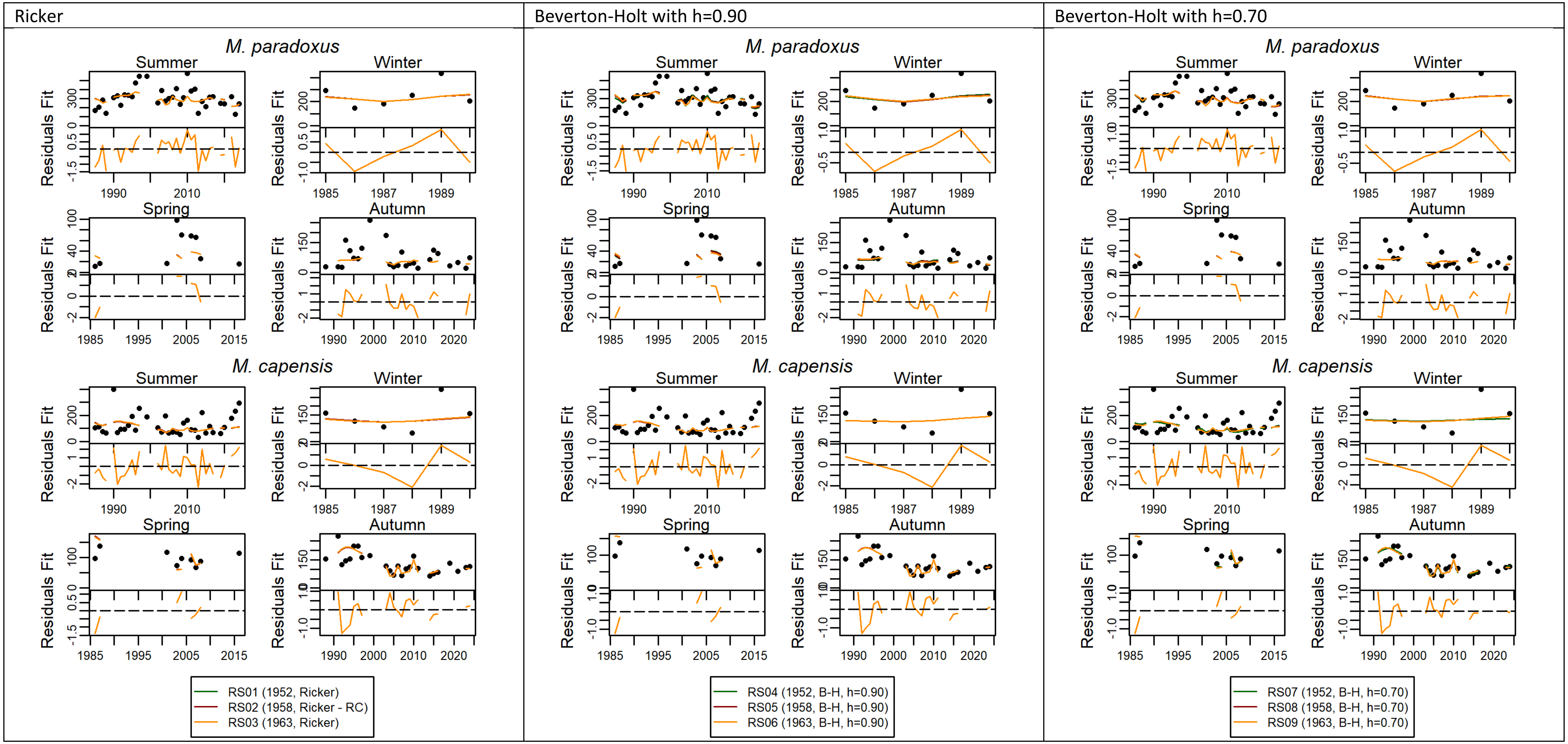


Figure B. 5: Fits to survey relative abundance indices are shown.

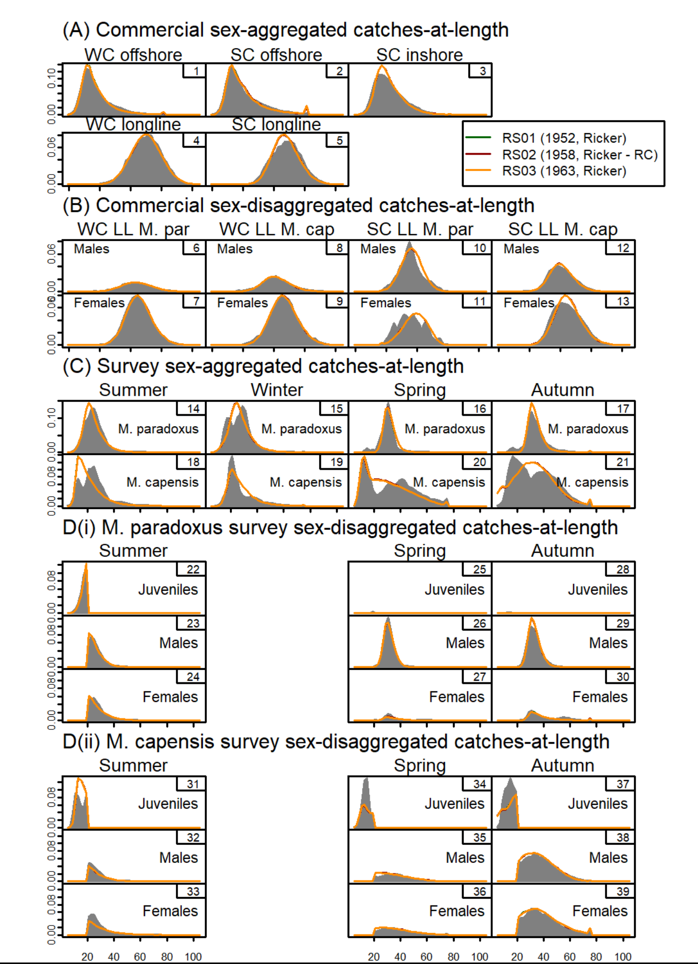
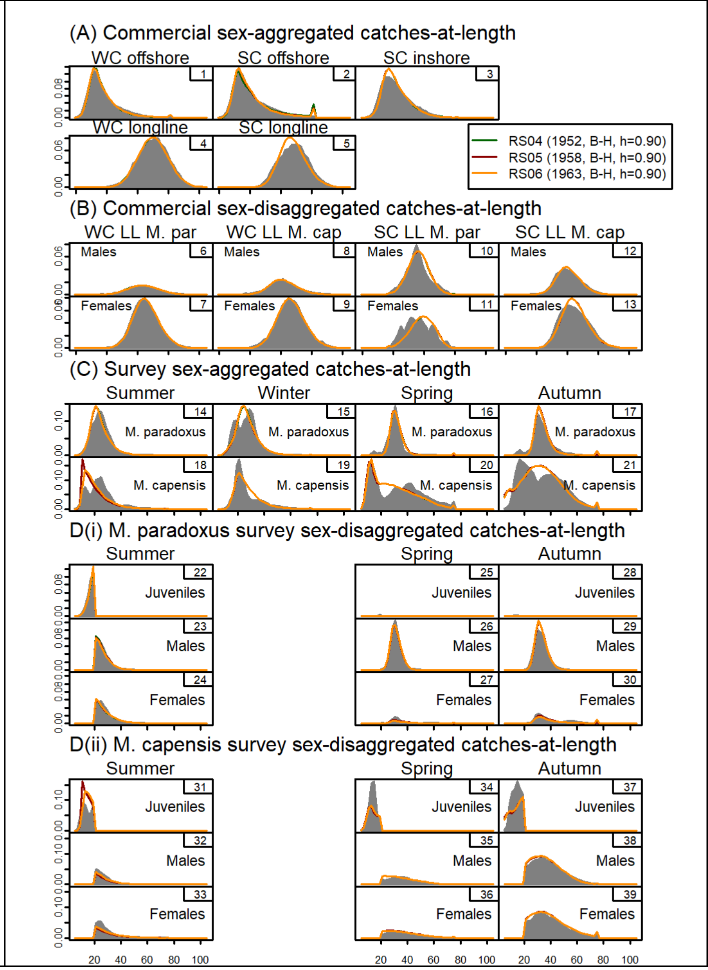
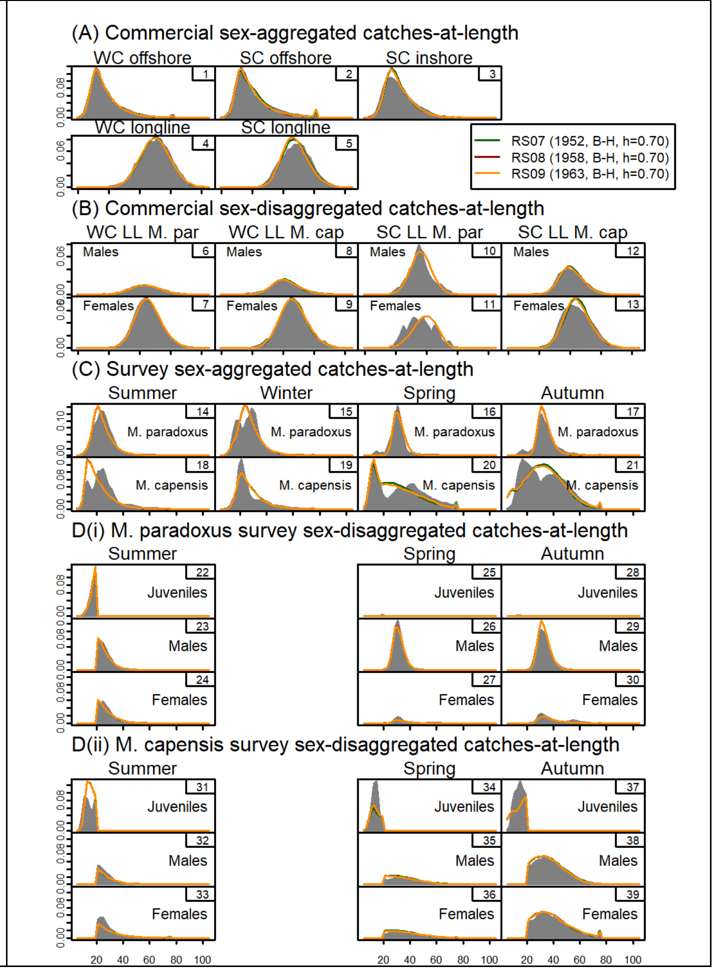
Ricker	Beverton-Holt with h=0.90	Beverton-Holt with h=0.70
		

Figure B. 6: Fits to commercial and survey catch-at-length data are shown.

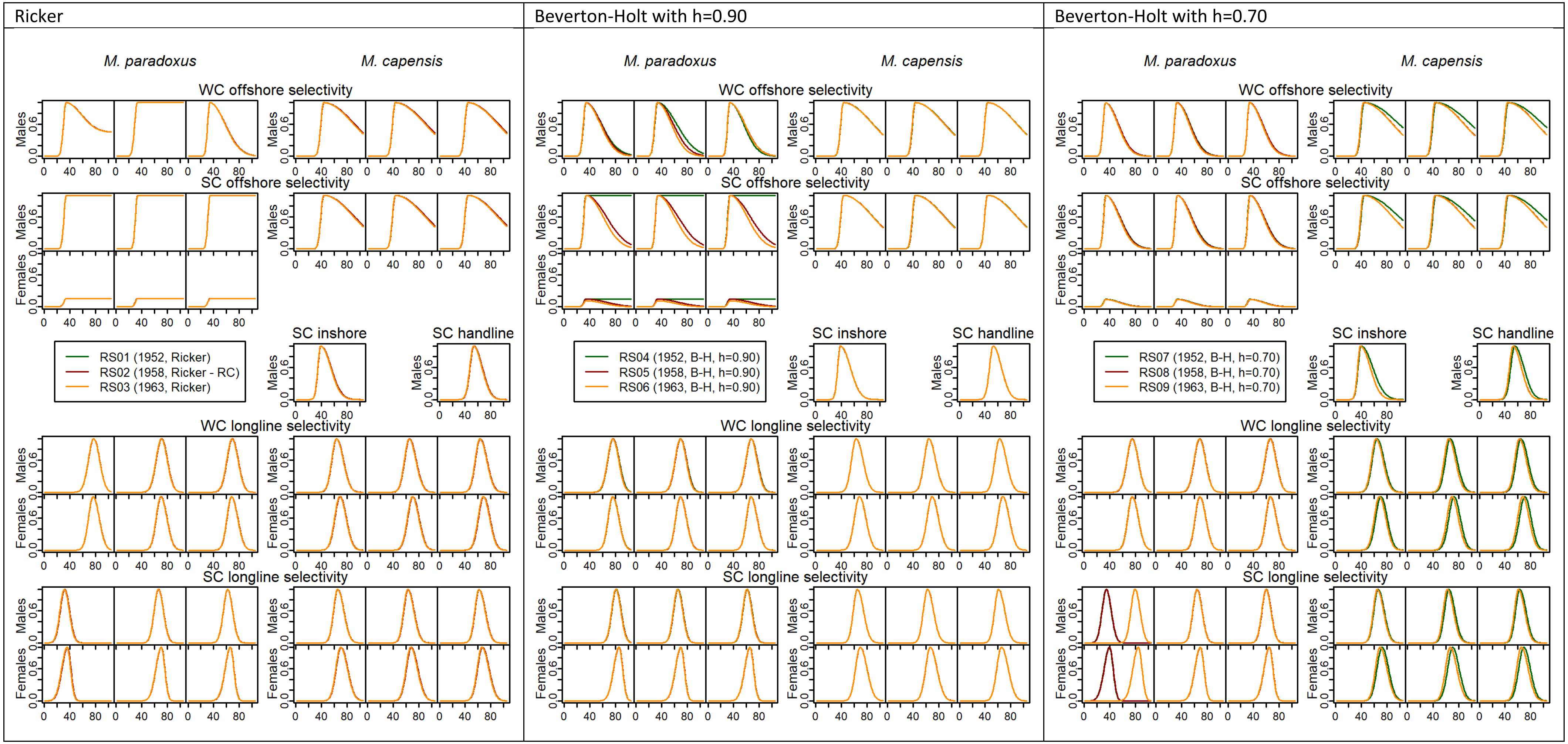


Figure B. 7a: The estimated **commercial selectivity-at-length** functions are shown in this Figure.

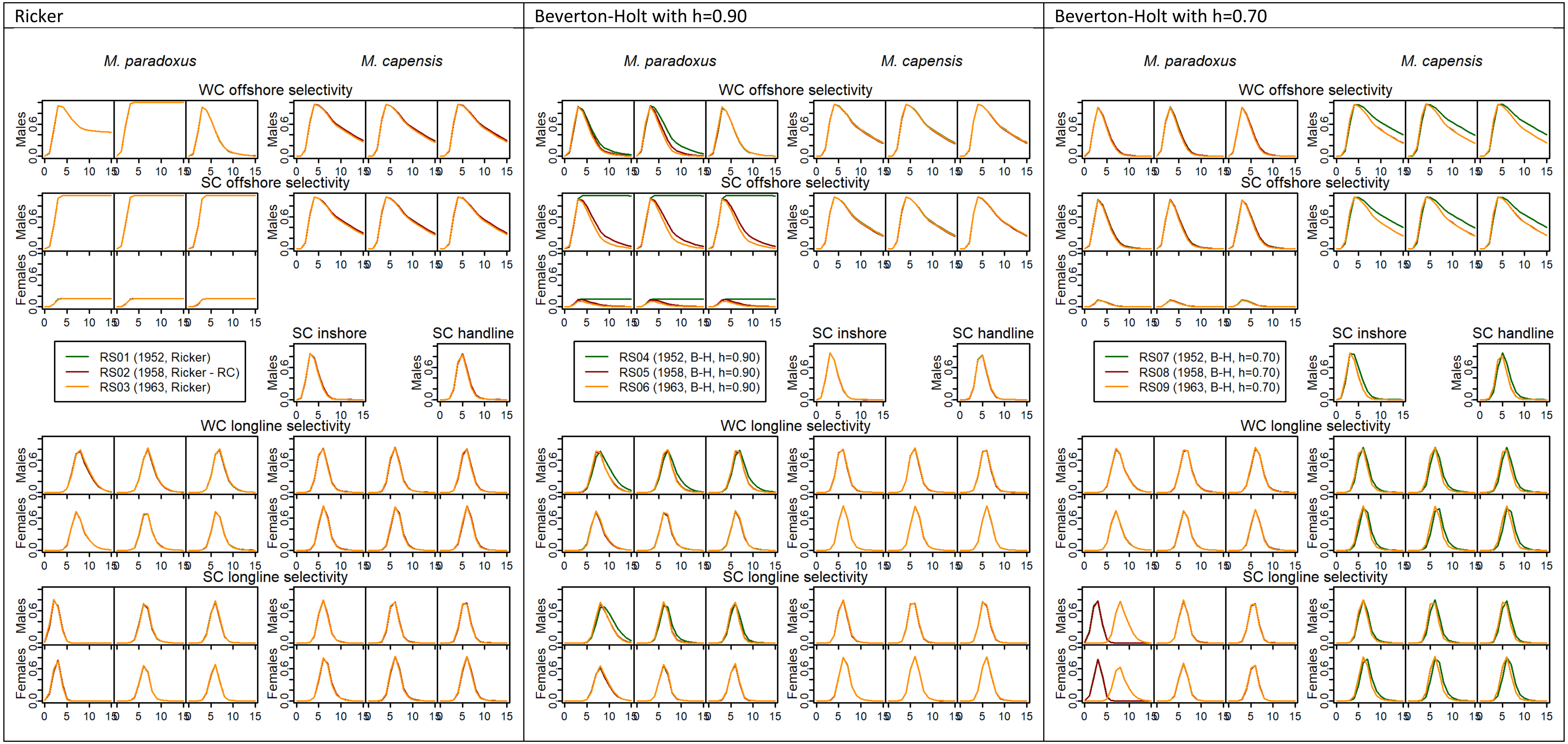


Figure B. 4b: The estimated **commercial selectivity-at-age** functions are shown in this Figure.

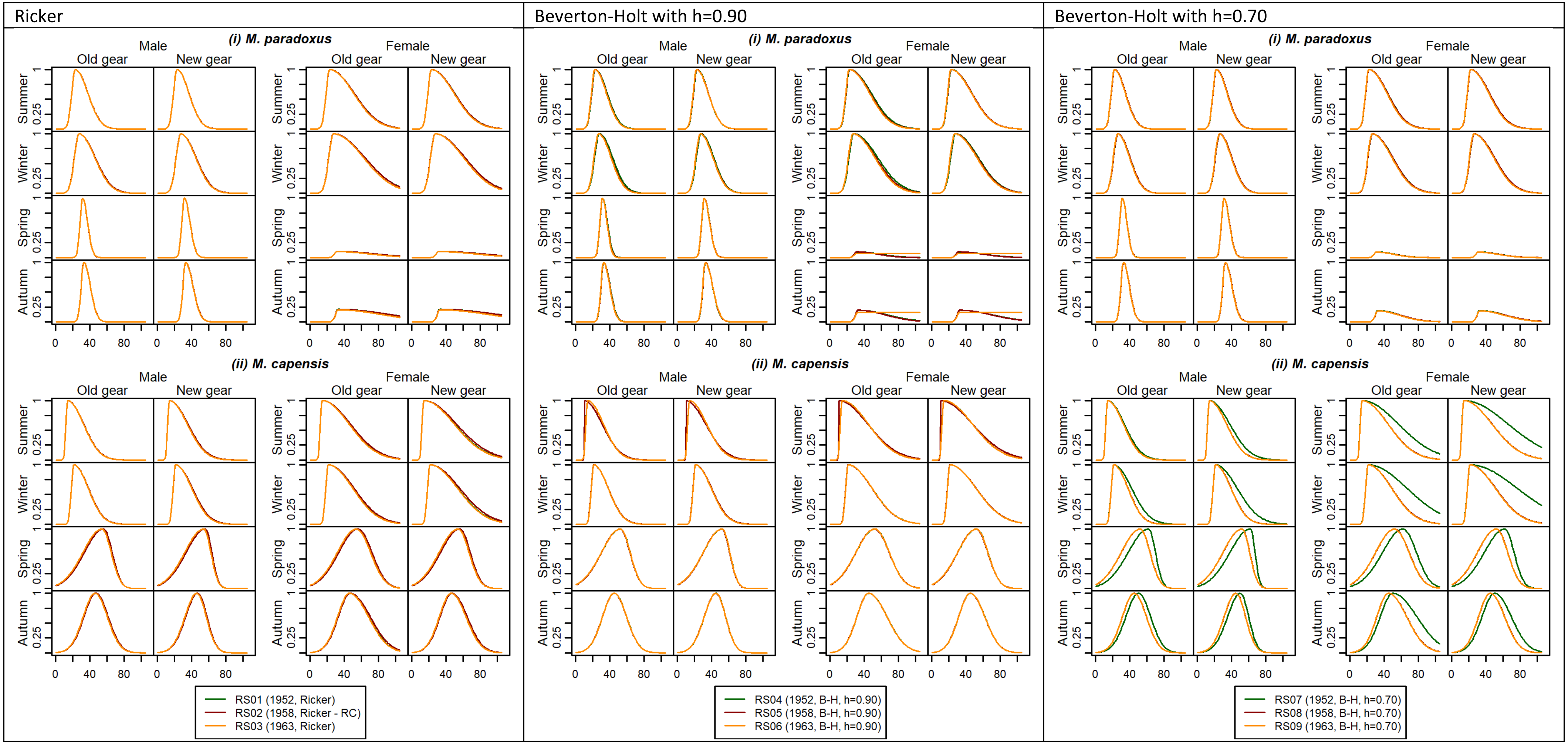


Figure B. 8a: Estimated **survey selectivity-at-length** functions are shown in this Figure.

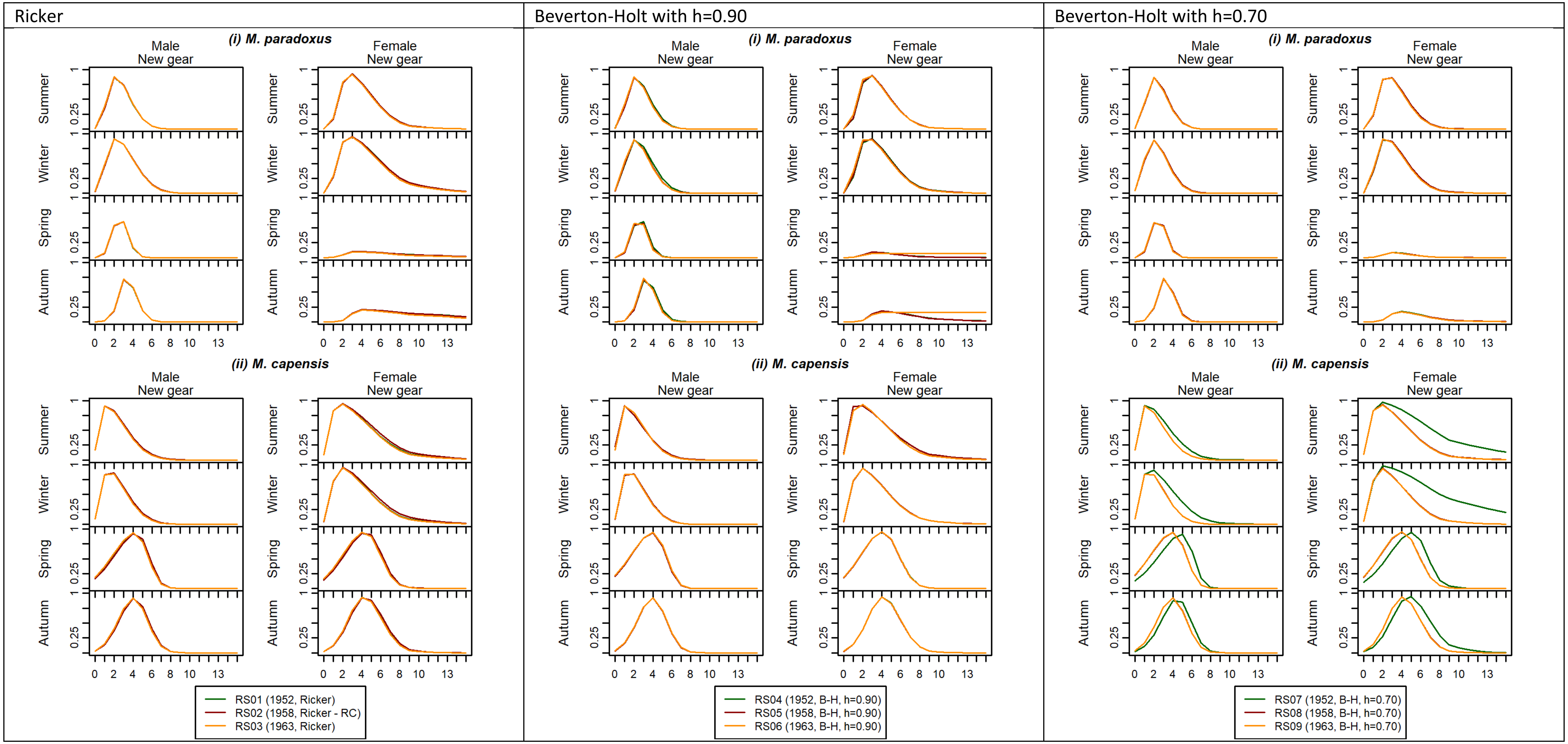


Figure B. 5b: Estimated **survey selectivity-at-age** functions are shown in this Figure. Selectivities are shown for the new gear only, as the old gear selectivity-at-age functions have not been coded into the standard assessment output at this point in time.

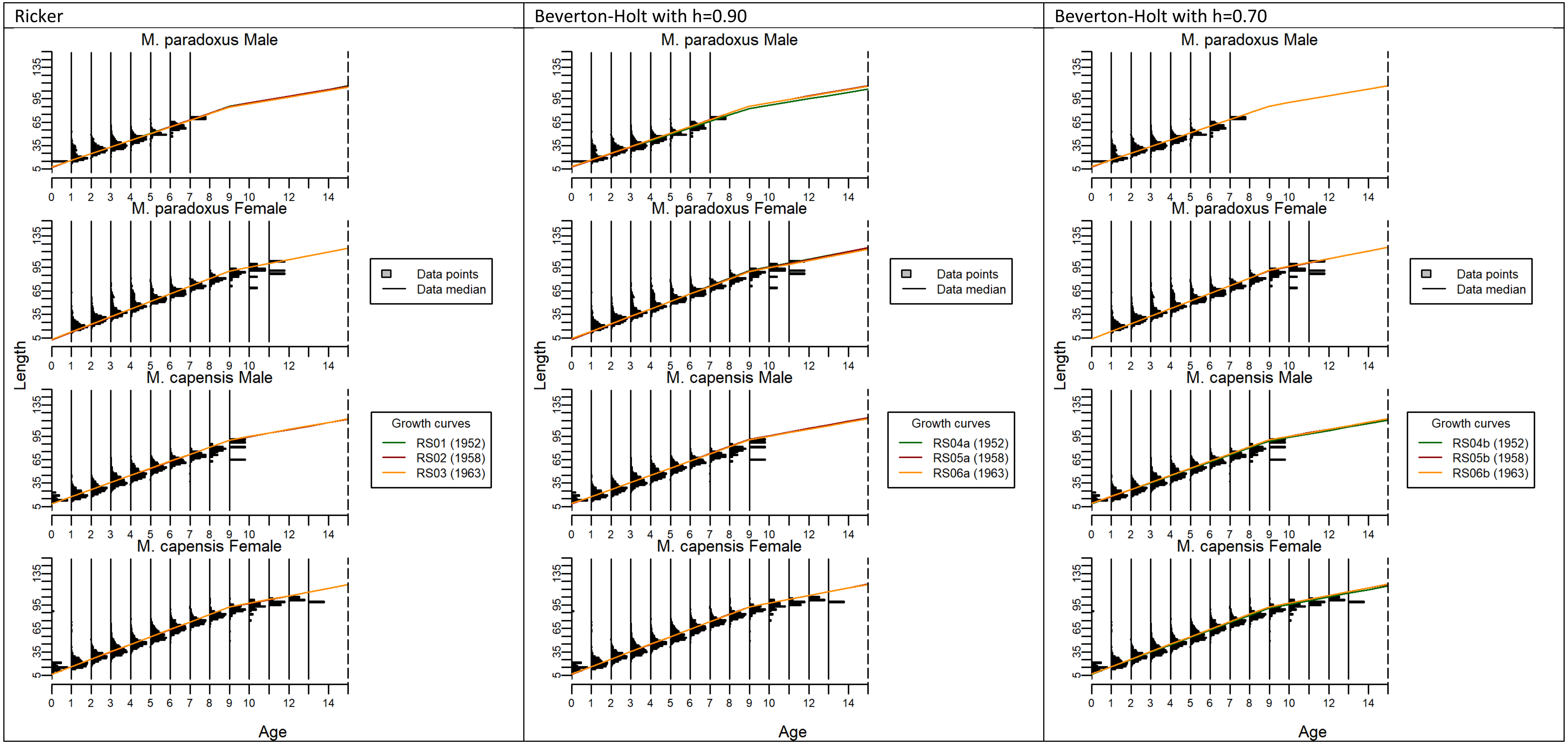


Figure B. 9: The estimated growth curves are superimposed on the length-at-age distributions derived from the age-length-key data.